

Robustness Analysis of Constructive Phase-Aligned PASS Placement Under Bounded Position Errors

Final Year Project Report

Abstract

Pinching-antenna systems (PASS) provide flexible antenna activation by pinching dielectric particles onto a waveguide, achieving single-user near-field focusing with a large effective aperture. Earlier studies on PASS have focused on nominal placement, rate maximization, antenna activation, and array gain under ideal pinching positions. This report turns to a different set of issues: when the constructive, phase-aligned PASS placement experiences bounded or random position errors, what degradation mechanisms dominate the response, which aspects remain analytically tractable, and which conclusions hold up under numerical verification.

Building on the PASS system model familiar from the array-gain literature, the report reformulates the constructive aligned placement as a feasible wavelength-snapped phase-alignment design that respects inter-element spacing constraints. It then derives a position-dependent phase sensitivity and a deterministic small-error lower bound for the normalized array gain.

For the studied symmetric setting with $N = 16$, $d = 3$ m, $f_c = 28$ GHz, $n_{\text{eff}} = 1.44$, and $\Delta_p = 0.5\lambda$, the first-order deterministic validity threshold is $\varepsilon/\lambda = 0.1713$, and within that region the deterministic lower bound is numerically indistinguishable from the box-best-found degradation curve. At the representative error level $\varepsilon/\lambda = 0.10$, the split-mode construction nearly matches the box-best-found degradation while common-bias motion remains almost harmless. The same second-order predictor also tracks the observed mean degradation across iid uniform, common-bias, and correlated Gaussian error models over the tested small-error range $0 \leq \varepsilon/\lambda \leq 0.10$. In the same-aperture baseline comparison studied here, the constructive aligned placement is stronger than a naive uniform-aperture design, but it is still outperformed by a best-found nominal-gain comparator.

The final conclusion is deliberately conservative. The present evidence supports a disciplined robustness characterization for one constructive PASS design in one studied symmetric configuration. It does not support any globally certified continuous-box optimum claim, any theorem that the split-mode pattern is the exact minimizer, or any broad design-optimality statement across PASS placements.

Keywords: pinching-antenna systems, PASS, position errors, robustness, bounded uncertainty, phase variance, near-field focusing

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“Treading upon iron caltrops, I ride without rest, not daring to look back.”

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I’m writing this to *American Football*. It isn’t happy, but it’s honest — like *La Haine*, it finds words for the thing that has nowhere else to go. I’ve been swimming a long time. Standing before knowledge, I feel grain-of-dust small — which isn’t unpleasant. At least it means I can still feel. More often there was struggle and self-doubt: *“Do I love wisdom enough? Do I love humanity enough?”* I asked, and never dared answer yes. I’m grateful that certain people waited on the shore and told me to keep going.

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Here, at the end of these words, another cycle begins. What we lived through wasn’t one act of great courage. It was a long series of ordinary endurances — every today held up by a yesterday-self who simply didn’t quit. And *Keep swimming until the sea turns blue*.

1 Introduction and Literature Survey

Pinching-antenna systems (PASS) were introduced as a flexible-antenna technology in which dielectric particles are pinched onto a waveguide so that radiation can be activated at arbitrary points along the guide [1]. In contrast to conventional fixed arrays, the same physical backbone can support different activation patterns without rebuilding the entire front-end. PASS therefore belongs to the broader family of flexible-antenna architectures alongside fluid antennas and movable antennas [2–5], but it also has its own modeling feature: the received phasor is affected not only by the free-space propagation distance between the user and the activated antenna element, but also by a guided-wave phase term along the dielectric waveguide. This dual phase structure is precisely what makes constructive near-field focusing possible, and it is also what makes PASS unusually sensitive to position perturbations.

1.1 Background and Motivation

This paper brings together two active research strands. The first involves near-field and large-aperture communications, where spherical-wave propagation, location-dependent phase, and aperture geometry are the essential modeling components [6]. The second strand treats flexible-antenna systems, considering the geometry of radiating elements as a controllable variable [2–5]. The pinching-antenna-systems (PASS) framework draws on both. Prior work on PASS has shown, for example, that downlink rates improve when activated positions are chosen appropriately [7]. The same flexibility also makes NOMA activation feasible [8] and can enforce uplink fairness constraints [9]. Separately, constructive alignment leads to a well-characterized nominal array gain [10]. More recent extensions have broadened the PASS scope toward sensing [11] and tutorial-style overviews [12]. However, existing studies continue to emphasize nominal activation, beam/rate optimization, and the expansion of application domains. A pressing question for physical deployment remains unanswered: what are the implications when the intended pinching positions cannot be hit exactly?

This is not a trivial practical matter. In an actual PASS system, activation points are realized by pinching or displacing dielectric elements to designated spots; precise placement is difficult to guarantee. Fabrication tolerances, placement offsets, actuator limits, and calibration errors all perturb those intended positions. Once a pinching position shifts, the free-space distance changes, the waveguide path length changes, and the coherent addition of the antenna responses deteriorates. Because the PASS phase term contains the effective refractive index n_{eff} , the sensitivity to position errors is amplified relative to purely free-space movable-antenna models. Recent movable-antenna literature has started to study the effect of position errors in near-field beamforming [13, 14], which makes the problem timely. Yet PASS deserves a dedicated treatment because its phase law is not simply geometric; it is the sum of a spherical-wave term and a guided-wave term. The resulting robustness story is therefore related to, but not identical with, the robustness story for conventional movable or fluid antennas.

Robustness also has to be decomposed carefully. A bounded independent tolerance on each

activated position, a coherent common drift of the whole aperture, and a correlated displacement field do not represent the same physical failure mode and should not be expected to produce the same degradation. A useful PASS robustness study must therefore do more than attach an error bar to nominal gain. A key task is to distinguish genuinely harmful error directions from those that are primarily common-mode, and to identify conditions under which a local analytical description can still be trusted. The body of this report follows that principle: bounded adversarial errors, stochastic perturbations, and baseline comparisons are developed as related but distinct lines of evidence.

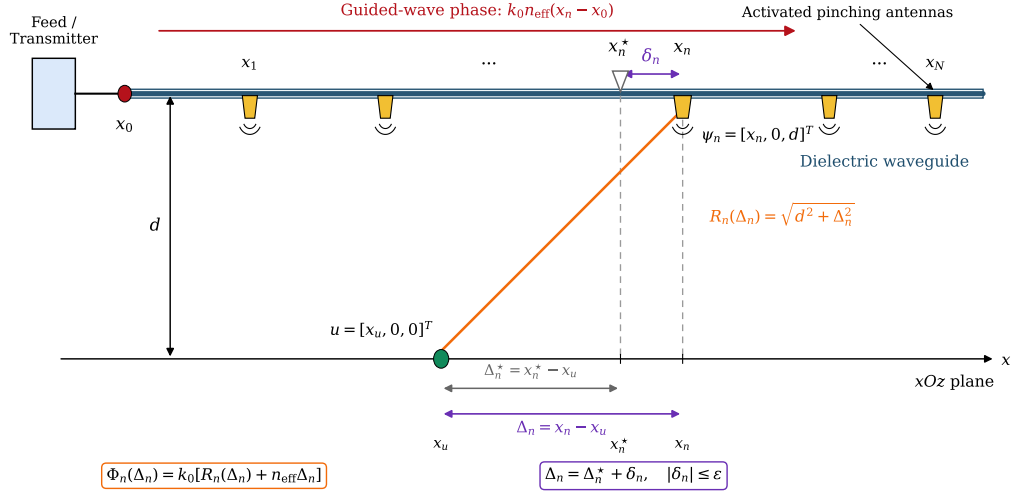


Figure 1: PASS system model used in this report, adapted from the array-gain formulation in [10]. The pinching antennas are activated along a dielectric waveguide, and the received phasor combines the free-space path from each activated point to the user with the guided-wave phase accumulated along the waveguide.

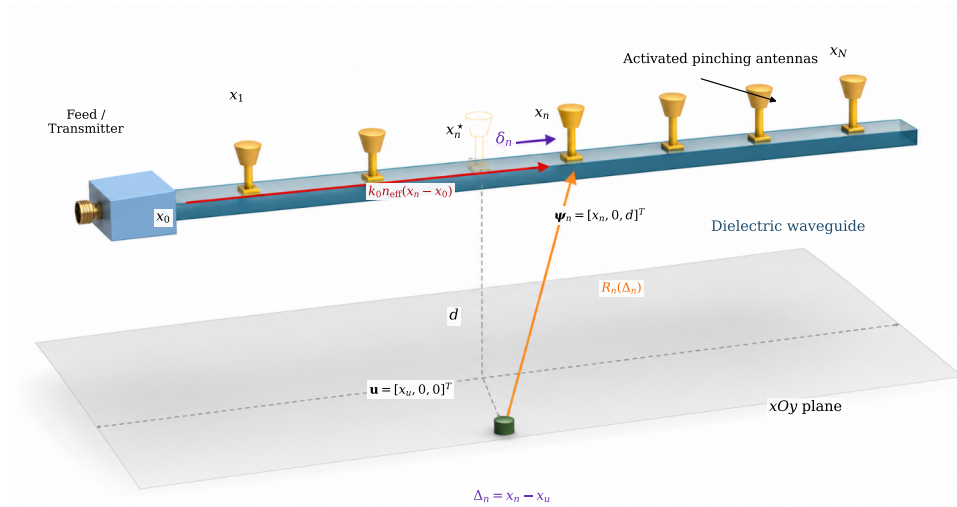


Figure 2: Three-dimensional view of the same PASS system model. This visual is included to make the waveguide, activated pinching antennas, user location, and propagation paths easier to read; the analytical model remains the two-dimensional symmetric geometry defined in Section 2.

1.2 Project Scope and Relation to the Base PASS Paper

This final-year project builds directly on the array-gain formulation of PASS in [10]. The starting point is the single-user symmetric geometry used in that paper, together with its constructive phase-alignment viewpoint. The base literature shows how to design a nominal placement so that the phasors add coherently under ideal pinching positions. This project keeps that nominal starting point fixed and studies a new problem: once the constructive phase-aligned feasible placement is perturbed by bounded or random position errors, which degradation mechanisms dominate, what part of the degradation can be bounded analytically, and how much of the remaining behavior can still be explained by a small set of interpretable quantities?

The distinction between inherited theory and the project’s own contribution remains explicit throughout the report. The system geometry, free-space path, waveguide phase, and normalized array-gain objective that define PASS are drawn from [10]. The constructive aligned placement adopted here reframes the phase-alignment concept as a wavelength-snapped design that satisfies spacing constraints—this reformulation constitutes the project-level contribution. With this feasible nominal placement in hand, the robustness analysis advances through four interconnected steps. A position-dependent phase sensitivity is derived to quantify how strongly each activated pinching position affects the coherent sum. Under bounded errors, a deterministic lower bound follows, valid within a first-order small-error regime. To account for the dominant error mechanism beyond that narrow bound, a weighted phase-variance perspective is introduced, explaining why a left–right split perturbation proves empirically damaging whereas a common signed shift remains almost harmless. The same phase-variance object is then extended to stochastic error models, producing a covariance-based second-order predictor that covers i.i.d., common-bias, and correlated perturbations.

Choosing to fix a single constructive nominal design rather than re-optimizing the placement under every uncertainty law is a deliberate methodological decision. It centers the investigation on deeply understanding one reproducible design instead of superficially comparing many partly incomparable designs. Such a strategy fits a long-form graduation report especially well: it makes the chain of evidence easier to follow, since the same geometry, nominal reference, and sensitivity profile persist from the model definition through the deterministic bound and the stochastic predictor to the baseline interpretation.

1.3 Literature Survey

The PASS literature can be read as progressing from architecture definition to nominal design and then to broader application scenarios. The foundational PASS papers establish the waveguide-and-pinching architecture and explain why movable activation points create a new form of geometry control in wireless systems [1, 15]. Building on that model, later studies examine what can be achieved when the pinching positions are assumed ideal: constructive array-gain analysis in the single-user setting [10], downlink rate-oriented design [7], NOMA-oriented activation strategies [8], and uplink fairness formulations [9]. Recent tutorial and outlook papers further widen

the scope toward system-level interpretation and sensing-oriented extensions [11, 12]. Taken together, these papers make the nominal potential of PASS increasingly clear, but they still leave the robustness of those nominal placements largely untreated.

What these PASS papers share is an ideal-actuation viewpoint. The technical questions differ, but once a placement or activation set is selected, the resulting coordinates are effectively treated as exact. That assumption is reasonable for demonstrating nominal capability, but it leaves two important questions open. First, it does not reveal whether constructive gain is naturally tolerant or intrinsically fragile. Second, it does not indicate which perturbation structures should be regarded as the relevant stress tests for PASS: independent per-element offsets, coherent drifts, or spatially correlated errors. The present report addresses those questions directly and uses them to separate deterministic, empirical, and stochastic statements later in the manuscript.

The broader flexible-antenna literature is also essential for positioning this project. Fluid antennas and movable antennas show that antenna geometry can be treated as a design variable rather than as a fixed hardware constraint [2–5]. This literature is useful in two ways. First, it provides the wider communications motivation for studying geometry adaptation in near-field and large-aperture regimes. Second, it clarifies the modeling rhythm that is now standard in this area: specify the geometry, state the mobility or placement constraint, define the channel law, and then optimize the performance objective induced by that controllable geometry. Even so, PASS cannot be reduced to a generic flexible-antenna model, because its phase law contains not only the spherical-wave term but also the guided-wave accumulation along the dielectric waveguide. That additional term is central to both constructive alignment and the position-sensitivity issue studied here.

The closest comparison class comes from recent robustness studies under geometry uncertainty. Near-field and large-aperture communications already show that location-dependent phase and spherical-wave propagation can make geometry errors much more consequential than in far-field beamforming [6]. More specifically, recent movable-antenna studies have started to quantify the effect of position errors in near-field beamforming and flexible-array optimization [13, 14]. These papers are important because they show that robustness cannot be treated as a minor implementation detail once antenna location becomes part of the design space. However, their uncertainty mechanism is still tied to free-space geometry control. The PASS setting adds the refractive-index-weighted waveguide term, so the relevant local sensitivity and the most damaging perturbation patterns need not match the movable-antenna case.

For that reason, the present survey is organized by technical axis rather than by author chronology. The key categories are PASS foundations and tutorials, PASS nominal design and application papers, the broader flexible-antenna context, and geometry-uncertainty robustness studies. Organizing the literature in this way suits a longer report, because it surfaces shared and missing assumptions simultaneously. In particular, it shows that the gap is not the broad statement that robustness has not been studied anywhere, but the narrower, more defensible claim that the cited literature does not adequately explain PASS-specific robustness for constructive phase-aligned placement.

These three strands of literature thus reveal a clear gap. The PASS community possesses well-developed nominal models and growing application coverage, and the wider flexible-antenna community has begun to confront geometry uncertainty; yet the cited PASS works still lack a focused explanation of how constructive phase-aligned PASS placement degrades under bounded and stochastic pinching-position errors. To fill this gap, the report limits itself to the single-user symmetric geometry of [10], a single feasible nominal design, and an investigation of how deterministic bounds, interpretable adversarial patterns, and stochastic covariance jointly explain robustness—a deliberate narrowness that separates deterministic small-error arguments from empirical observations instead of blending them into a broader, weaker claim.

By condensing the literature architecture into the four categories that matter here, Table 1 serves both as a scannable survey chapter and as a map that situates the project precisely within existing PASS and flexible-antenna work.

Table 1: Category-based literature map and the specific gap addressed in this report.

Category	Representative focus	Key references	Relevance to this report
PASS foundations and architecture	Waveguide-enabled activation, system concept, and broad PASS framing	[1, 12, 15]	Establishes the PASS model class, but does not analyze robustness under pinching-position errors.
PASS nominal design and applications	Nominal constructive gain, downlink rate design, NOMA activation, uplink fairness, sensing extensions	[7–11]	Provides the nominal starting point and application context inherited here.
Flexible-antenna context	Fluid and movable antennas as geometry-controlled communication systems	[2–5]	Explains why geometry control matters, but does not capture the PASS-specific guided-wave phase term.
Position uncertainty and near-field robustness	Geometry-error sensitivity in near-field or movable-antenna beamforming	[6, 13, 14]	Gives the closest robustness analogue and motivates a dedicated PASS error study.

In compact form, the inherited material from prior PASS work is the system geometry, channel law, and nominal phase-alignment motivation. The new material in this project is the feasible constructive placement reformulation, the bounded- and stochastic-error robustness analysis, and

the narrow same-aperture baseline comparison. What is not claimed is any exact continuous-box worst-case solution or any globally optimal PASS placement result.

1.4 Objectives, Contributions, and Non-Claims

The technical gap addressed here can be stated precisely. Existing PASS literature explains how constructive phase alignment improves nominal coherent addition [10, 15]. What is still missing is a position-error theory that is (1) local enough to permit explicit derivations and validity conditions, (2) sufficiently interpretable to explain why distinct perturbation patterns produce qualitatively different behaviour, (3) broad enough to bridge deterministic bounded-error models with stochastic error covariance models, and (4) conservative enough to avoid over-claiming global optimality or universal design rules.

This report addresses that gap for one coherent fixed configuration. The simulation setup is $N = 16$, $d = 3$ m, $f_c = 28$ GHz, $n_{\text{eff}} = 1.44$, and $\Delta_p = 0.5\lambda$. The constructive phase-aligned placement generated from this setup is then analyzed under bounded box errors, Monte Carlo perturbations, and several placement baselines. The result is a complete evidence chain from system model to design construction, perturbation theory, numerical validation, and claim-disciplined conclusions.

The main contributions of the report are summarized below.

- A PASS robustness formulation is built directly on the array-gain model of [10], but rewritten around the constructive phase-aligned feasible placement generated under spacing constraints. This makes the nominal design explicit and reproducible.
- A position-dependent phase sensitivity ξ_n and a deterministic lower bound are derived for bounded position errors. For the studied design, the first-order deterministic validity threshold is $\varepsilon/\lambda = 0.1713$, and within that region the lower bound tracks the box-best-found degradation curve almost exactly.
- A weighted phase-variance interpretation is developed to unify deterministic and stochastic robustness behavior. The same object explains why an empirical left-right split-mode perturbation nearly matches the box-best-found adversary before the first null, while common-bias motion produces almost no degradation.
- A covariance-based stochastic predictor is derived and validated for iid uniform, common-bias, and correlated Gaussian error models. This gives a compact explanation of why statistically smooth perturbations are much less damaging than independent ones with the same marginal variance.
- A narrow comparative design statement is established: in the tested same-aperture comparison, the constructive aligned placement is stronger than a naive uniform-aperture placement, especially in box-best-found robustness, but it is not the best tested design because a best-found nominal-gain comparator outperforms it on every reported metric.

Taken together, these contributions form a single, continuous argument—not five independent points. The constructive placement provides the reference geometry; the deterministic analysis establishes the bounds of what can be justified in the small-error regime. The split-mode mechanism explains the dominant harmful bounded perturbation up to the first null, the covariance predictor captures the ordering of the stochastic curves, and the baseline section measures the design value retained by the constructive rule relative to straightforward same-aperture alternatives. The logic constructed at this stage is what later allows the conclusions to remain conservative without becoming vague.

The report also explicitly disclaims several assertions. It does not claim a globally certified continuous-box optimum for the worst-case problem, nor that the split-mode pattern is the exact optimizer of the bounded-box adversary. No assertion of general design optimality is made over arbitrary PASS placements, antenna counts, or geometries. Further, the stochastic predictor is not extended beyond the tested small-error range or the constructive design examined here. These disclaimers are integral to the argument, not incidental; they shape how the evidence is presented.

1.5 Report Organization

We structure the rest of the paper accordingly. Section 2 outlines the framework, covering the inherited PASS model, constructive nominal placement, the position-error formulation, a deterministic lower bound, the weighted phase-variance mechanism, and the numerical evaluation protocol. Section 3 reports the results of deterministic, stochastic, and baseline-comparison analyses. In Section 4, we analyze the mechanisms behind these results and disentangle the deterministic small-error guarantees from the empirical observations. Section 5 concludes with key takeaways and promising avenues for future research. The appendices provide the full derivations and secondary sensitivity analyses that support the main text.

2 Methods

The technical contract of this report is defined in this section. We take the inherited PASS system model as a starting point and then introduce three elements: the constructive nominal placement used throughout the study, the bounded and stochastic formulations of position error, and the numerical workflow that generates the evidence discussed later. A central concern is to maintain a clear separation among exact definitions, local approximations, and the best-found numerical procedures.

2.1 PASS System Model and Constructive Nominal Placement

2.1.1 Base PASS Geometry and Channel Model

The system model adopted in this report is the single-user PASS geometry analyzed in [10]. A dielectric waveguide is aligned with the x axis at height d , a single-antenna user is located at $\mathbf{u} = [x_u, 0, 0]^T$, and N pinching antennas are activated on the guide at positions x_n , with N even and indices $n \in \{-N/2, \dots, -1, 1, \dots, N/2\}$. Following [10], the position of the n th activated pinching antenna is

$$\boldsymbol{\psi}_n = [x_n, 0, d]^T, \quad \Delta_n \triangleq x_n - x_u, \quad R_n(\Delta_n) = \|\mathbf{u} - \boldsymbol{\psi}_n\| = \sqrt{d^2 + \Delta_n^2}. \quad (1)$$

The free-space path contributes the usual spherical-wave phase $k_0 R_n(\Delta_n)$ and amplitude factor $1/R_n(\Delta_n)$. What distinguishes PASS from a purely free-space movable-antenna model is the guided-wave phase accumulated along the dielectric waveguide. If x_0 denotes the feed-point coordinate and n_{eff} is the effective refractive index, then the effective complex channel from the n th activated pinching antenna to the user can be written as [10]

$$h_n(\Delta_n) = \frac{\eta^{1/2}}{R_n(\Delta_n)} e^{-jk_0 R_n(\Delta_n)} e^{-jk_0 n_{\text{eff}}(x_n - x_0)}, \quad k_0 = \frac{2\pi}{\lambda}. \quad (2)$$

The received baseband signal with equal power allocation across the activated antennas is

$$y = \sqrt{\frac{P}{N}} \sum_n h_n(\Delta_n) s + z, \quad z \sim \mathcal{CN}(0, \sigma^2), \quad (3)$$

and the base PASS array-gain quantity used throughout this report is

$$a(\Delta) \triangleq \frac{\text{SNR}}{P/\sigma^2} = \frac{\eta}{N} \left| \sum_n \frac{e^{-j\Phi_n(\Delta_n)}}{R_n(\Delta_n)} \right|^2, \quad \Phi_n(\Delta_n) = k_0(R_n(\Delta_n) + n_{\text{eff}}\Delta_n), \quad (4)$$

after absorbing the feed-point offset into a common reference phase. Equation (4) is the key object inherited from [10]. It already makes clear why PASS differs from a standard movable-antenna model: the phase changes with both $R_n(\Delta_n)$ and $n_{\text{eff}}\Delta_n$. Since $n_{\text{eff}} > 1$ in the present setup, small pinching-position errors can produce larger phase changes than those induced by free-space geometry alone.

Equation (4) also shows why nominal performance and robustness should not be treated as

identical questions. A placement can align the phasors well at the operating point while still being locally sensitive to perturbation if the phase derivatives vary strongly across the aperture. The later baseline section is included partly for that reason: it checks whether a placement that looks attractive at nominal alignment also remains attractive once the same aperture is stressed by position uncertainty.

For manuscript clarity, the report uses two closely related gain quantities. The *raw array-gain value* of a placement is the raw quantity in (4), and this is the scale used in the same-aperture baseline comparison. When the question is how one fixed nominal design degrades under perturbation, Sections 2.2–4.2 instead use a design-referenced normalized gain. Let

$$\phi_n^* \triangleq \Phi_n(\Delta_n^*), \quad a_{\text{ref}}(\Delta; \Delta^*) \triangleq \frac{\eta}{N} \left| \sum_n \frac{e^{-j(\Phi_n(\Delta_n) - \phi_n^*)}}{R_n(\Delta_n)} \right|^2, \quad (5)$$

and define

$$\tilde{a}(\Delta; \Delta^*) \triangleq \frac{a_{\text{ref}}(\Delta; \Delta^*)}{a_{\text{ideal}}(\Delta^*)}, \quad a_{\text{ideal}}(\Delta^*) \triangleq \frac{\eta}{N} \left(\sum_n \frac{1}{R_n(\Delta_n^*)} \right)^2. \quad (6)$$

By construction, $\tilde{a}(\Delta^*; \Delta^*) = 1$. The design-referenced normalized gain is plotted in the perturbation-focused results of Sections 3.1–3.2. Section 3.3 then adopts raw array-gain values for cross-placement comparison. To simplify the subsequent perturbation formulas, $\tilde{a}$ is often denoted simply as the *normalized gain* once the reference design Δ^* is fixed.

The raw-normalized split is not a notational artifact. It serves two different analytical purposes. When a single placement is perturbed, the relevant question is how much of its coherent-gain potential is preserved; hence the design-referenced normalized gain is the appropriate measure. When comparing different placements, however, design-specific normalization would hide nominal-performance disparities and make a weaker design appear artificially strong after rescaling. For this reason, the report uses $\tilde{a}$ to characterise perturbations and raw array gain for cross-placement comparisons, and explicitly reminds the reader of this distinction wherever figures or tables might be misconstrued.

2.1.2 Constructive Phase-Aligned Feasible Placement

The nominal design studied here is not a globally solved optimizer of (4). Instead, it is a constructive feasible placement that enforces approximate phasor alignment under the same minimum-spacing constraint used in the numerical study:

$$|\Delta_n - \Delta_{n'}| \geq \Delta_p \lambda, \quad \Delta_{-n} = -\Delta_n. \quad (7)$$

The phase-alignment idea is inherited from [10]. If all phasors in (4) share the same phase modulo 2π , then the coherent sum approaches the triangle-inequality envelope. This suggests the alignment condition

$$\Phi_n(\Delta_n) = C + 2\pi m_n, \quad m_n \in \mathbb{Z}, \quad (8)$$

which is equivalent to

$$R_n(\Delta_n) + n_{\text{eff}}\Delta_n = C' + m_n\lambda, \quad C' \triangleq \frac{C}{k_0}. \quad (9)$$

This condition does not by itself produce a feasible placement, because the integers m_n and the common offset C' must still respect symmetry and minimum spacing. The project therefore adopts the following constructive procedure. First, the positive-side coordinates are initialized on a minimum-spacing grid,

$$\Delta_n^{(0)} = \frac{\Delta_p\lambda}{2} + (n-1)\Delta_p\lambda, \quad n = 1, \dots, \frac{N}{2}. \quad (10)$$

Second, each candidate position is converted into a wavelength-snapped target distance

$$d_n = \lambda \left\lceil \frac{\sqrt{d^2 + (\Delta_n^{(0)})^2} + n_{\text{eff}}\Delta_n^{(0)}}{\lambda} \right\rceil, \quad (11)$$

and the aligned coordinate is obtained by solving

$$\sqrt{d^2 + \Delta_n^2} + n_{\text{eff}}\Delta_n = d_n. \quad (12)$$

In the numerical study, (12) is solved by a bracketed one-dimensional root finder, but the same equation also yields the quadratic form

$$(n_{\text{eff}}^2 - 1)\Delta_n^2 - 2d_n n_{\text{eff}}\Delta_n + (d_n^2 - d^2) = 0, \quad (13)$$

whose physically feasible root is

$$\Delta_n = \frac{d_n n_{\text{eff}} - \sqrt{d_n^2 + d^2(n_{\text{eff}}^2 - 1)}}{n_{\text{eff}}^2 - 1}, \quad n_{\text{eff}} \neq 1. \quad (14)$$

If the resulting coordinate violates the minimum-spacing constraint with the previously accepted element, the wavelength target is incremented by one more λ and the solve is repeated. The negative-side positions are then completed by symmetry. The resulting constructive placement is denoted by Δ^* .

The integers m_n in (8) can be interpreted as alignment classes separated by full wavelengths of combined free-space and guided-wave path. That interpretation is useful because it turns the design from a black-box coordinate search into a sequence of physically meaningful wavelength choices. The constructive algorithm used here does not claim to explore those choices globally. Instead, it selects the nearest feasible aligned targets compatible with symmetry and minimum spacing, which is exactly the level of design discipline needed for a reproducible robustness study.

The key point is that Δ^* is a *constructive phase-aligned feasible placement*. It is designed to make the nominal coherent sum large, but it is not presented as a global optimizer of (4). This distinction matters later when robustness is compared with other placements. The report does not rely on a nominal optimality theorem; it relies only on the fact that the chosen constructive design

is explicitly generated, reproducible, and exactly the design used by the robustness experiments.

2.2 Position-Error Model, Phase Sensitivity, and Deterministic Analysis

Once the nominal design Δ^* is fixed, the actual pinching positions are modeled as

$$\Delta_n = \Delta_n^* + \delta_n, \quad |\delta_n| \leq \varepsilon. \quad (15)$$

The worst-case bounded-error performance of the chosen design is therefore

$$a_{\text{wc}}(\Delta^*, \varepsilon) = \min_{\|\delta\|_\infty \leq \varepsilon} a(\Delta^* + \delta). \quad (16)$$

Because the objective is nonconvex and highly oscillatory, (16) is not solved globally in this report. In the perturbation study, the manuscript reports its design-referenced normalized counterpart

$$\tilde{a}_{\text{wc}}(\Delta^*, \varepsilon) = \min_{\|\delta\|_\infty \leq \varepsilon} \tilde{a}(\Delta^* + \delta; \Delta^*), \quad (17)$$

because the reference design is fixed throughout Sections 3.1–3.2. The numerical analysis evaluates that quantity using a continuous-box adversary labeled *box-best-found*, which combines exhaustive corner enumeration with box-constrained local refinement. Throughout the report we maintain the distinction between (16) as the raw array-gain definition, (17) as its reported normalized version, and the box-best-found solver as an implementation-level detail.

We select the ℓ_∞ box model because it supplies a clean, interpretable notion of tolerance: each activated position may deviate independently by at most ε . Other uncertainty descriptions are feasible, but this choice forms a natural entry point for actuator or placement tolerances and permits a clear separation of common-mode and differential motion. A coherent drift of all positions represents one distinguished direction within the box; a split perturbation represents another. The bounded-error question pursued here is thus not simply “how large the error can become,” but “which directions, under the identical tolerance budget, most severely degrade coherence.”

The first analytical quantity needed for robustness is the phase sensitivity at the nominal design. Differentiating the phase in (4) gives

$$\Phi'_n(\Delta_n) = k_0 \left(\frac{\Delta_n}{\sqrt{d^2 + \Delta_n^2}} + n_{\text{eff}} \right). \quad (18)$$

Evaluated at the constructive nominal placement, the dimensionless sensitivity is

$$\xi_n \triangleq \frac{\Delta_n^*}{\sqrt{d^2 + (\Delta_n^*)^2}} + n_{\text{eff}} = \sin \theta_n^* + n_{\text{eff}}, \quad (19)$$

where θ_n^* is the nominal angle seen from the user to the n th pinching antenna. The local phase error induced by a small position perturbation is then

$$\Delta\phi_n \approx k_0 \xi_n \delta_n. \quad (20)$$

This simple expression already explains why PASS is more fragile than a free-space-only movable array. In a conventional movable-antenna model, the first-order coefficient is essentially $\sin \theta_n^* \in (-1, 1)$. In PASS, the term is shifted by n_{eff} , so the whole sensitivity range is centered around the refractive index rather than around zero. For the present design,

$$\xi_n \in [1.4203, 1.4597], \quad \xi_{\max} = 1.4597. \quad (21)$$

Since all ξ_n are positive, any common signed shift rotates all phasors by nearly the same amount; an adversarial perturbation that degrades performance must therefore introduce relative phase dispersion rather than merely a global rotation. In the symmetric case considered here, the amplitude-weighted mean sensitivity (defined later) equals 1.44.

The sensitivity profile encapsulates both the geometric and guided-wave physics in a single local scalar. The term $\sin \theta_n^*$ encodes the geometric contribution of the user-to-element angle, while the additive n_{eff} term represents the guided-wave contribution shared by the whole aperture. The narrow range shown in (21) thus admits a clear interpretation: most of the PASS-specific sensitivity behaves as a common offset, and the small differential variation about that offset is what governs coherent loss. This observation later identifies the centered weighted phase variance—rather than the absolute phase shift alone—as the correct mechanism variable.

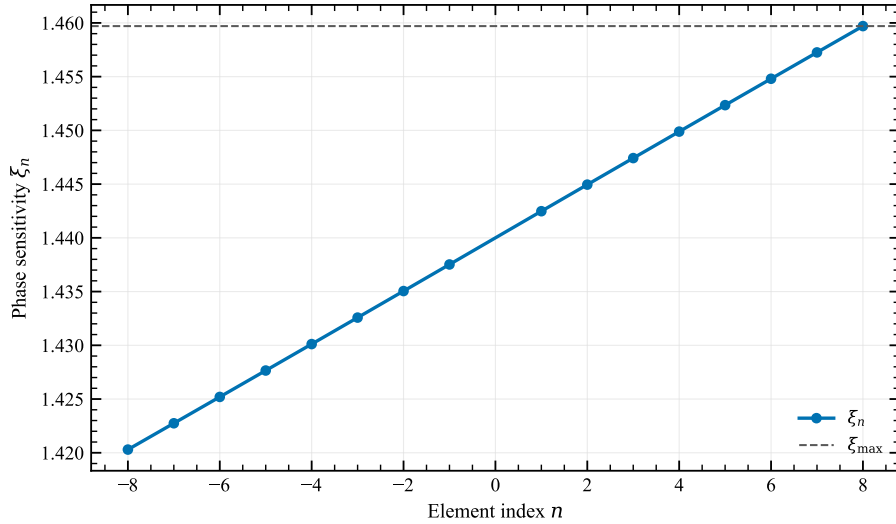


Figure 3: Distribution of the first-order phase sensitivity ξ_n over the constructive aligned placement. The small variation around $n_{\text{eff}} = 1.44$ is the key to both the split-mode mechanism and the near-harmlessness of common-bias motion.

To obtain a deterministic small-error lower bound, define the normalized coherent sum

$$\rho(\delta) \triangleq \frac{\sum_n R_n(\Delta_n^*)^{-1} e^{-j(\Phi_n(\Delta_n^* + \delta_n) - \Phi_n(\Delta_n^*))}}{\sum_m R_m(\Delta_m^*)^{-1}}. \quad (22)$$

If the per-element phase deviation stays within $[-\pi/2, \pi/2]$, then each phasor contributes a nonnegative real projection. A conservative amplitude-aware lower bound follows:

$$\tilde{a}(\Delta^* + \delta; \Delta^*) \geq \left(\frac{\sum_n \frac{1}{\sqrt{d^2 + (|\Delta_n^*| + \varepsilon)^2}} \cos(k_0 |\xi_n| \varepsilon)}{\sum_m \frac{1}{R_m(\Delta_m^*)}} \right)^2. \quad (23)$$

The crucial validity condition is

$$k_0 |\xi_n| \varepsilon \leq \frac{\pi}{2} \quad \text{for all } n, \quad (24)$$

which produces the first-order validity threshold

$$\varepsilon_{\text{valid}} = \frac{\pi/2}{k_0 \xi_{\text{max}}}, \quad \frac{\varepsilon_{\text{valid}}}{\lambda} = 0.1712677618. \quad (25)$$

Since all ξ_n are positive, a common signed shift rotates all phasors in nearly the same direction; conversely, a harmful adversarial pattern must introduce *relative* phase spread, not merely a global rotation. In the symmetric case under study, the amplitude-weighted mean sensitivity defined later takes the value 1.44.

The sensitivity profile also condenses the geometry and waveguide physics into a single local scalar. The $\sin \theta_n^*$ term captures the geometric contribution from the user-to-element angle, while the additive n_{eff} accounts for the guided-wave component common to the entire aperture. The narrow spread of ξ_n in (21) therefore has a direct interpretation: most of the PASS-specific sensitivity acts as a common offset, and the coherent loss depends on the small differential variation around that offset. This observation underlies the later choice of centered weighted phase variance—rather than absolute phase shift—as the relevant mechanism variable.

2.3 Weighted Phase Variance and Covariance-Based Stochastic Predictor

The deterministic lower bound is useful but incomplete. It explains small-error degradation within its validity condition, but it does not explain why the box-best-found adversary is harmful or why different stochastic error models behave so differently. For that, the project adopts a weighted phase-variance viewpoint.

Define normalized amplitude weights at the nominal placement by

$$\alpha_n = \frac{R_n(\Delta_n^*)^{-1}}{\sum_m R_m(\Delta_m^*)^{-1}}, \quad \sum_n \alpha_n = 1, \quad (26)$$

and the normalized coherent sum

$$\rho(\delta) = \sum_n \alpha_n e^{-jk_0 \xi_n \delta_n}. \quad (27)$$

Expanding the exponential for small phase errors gives

$$e^{-jk_0 \xi_n \delta_n} = 1 - jk_0 \xi_n \delta_n - \frac{1}{2} k_0^2 \xi_n^2 \delta_n^2 + O(\|\delta\|^3). \quad (28)$$

Let $\mu_\phi = \sum_n \alpha_n k_0 \xi_n \delta_n$. Then, up to second order,

$$|\rho(\delta)|^2 \approx 1 - \sum_n \alpha_n (k_0 \xi_n \delta_n)^2 + \left(\sum_n \alpha_n k_0 \xi_n \delta_n \right)^2 = 1 - \text{Var}_\alpha(k_0 \xi_n \delta_n). \quad (29)$$

Equation (29) is the central mechanism equation of the report. At second order, normalized gain loss is controlled not by a common phase rotation but by the *weighted variance* of the element-wise phase errors. A perturbation that moves every activated pinching position in almost the same effective phase direction produces very little degradation, whereas a perturbation that maximizes the spread of the weighted phase profile collapses the coherent sum.

This immediately motivates the empirical split-mode construction. Let $\bar{\xi}_\alpha = \sum_n \alpha_n \xi_n$. A natural box-feasible candidate that enlarges the centered phase spread is

$$\delta_n^{\text{split}} = \varepsilon \text{sign}(\xi_n - \bar{\xi}_\alpha). \quad (30)$$

In the current symmetric geometry, the weighting is symmetric and $\bar{\xi}_\alpha = n_{\text{eff}}$, so

$$\delta_n^{\text{split}} = \varepsilon \text{sign}(\sin \theta_n^*) = \varepsilon \text{sign}(\Delta_n^*), \quad (31)$$

which is exactly an outward left-right split. This pattern is not claimed to be the exact minimizer of the continuous-box problem. It is only a constructive, interpretable, and box-feasible adversarial candidate. The numerical results later show that it tracks the box-best-found curve almost perfectly before the first null.

The same phase-variance viewpoint also yields a compact stochastic predictor. Let the error vector be zero mean with covariance $\Sigma_\delta = \mathbb{E}[\delta \delta^T]$. Define

$$M_\alpha = \text{Diag}(\alpha) - \alpha \alpha^T, \quad D_\xi = \text{Diag}(\xi). \quad (32)$$

Taking expectation in (29) leads to

$$\mathbb{E}[\tilde{a}(\Delta^* + \delta; \Delta^*)] \approx 1 - k_0^2 \text{tr}(M_\alpha D_\xi \Sigma_\delta D_\xi). \quad (33)$$

Equation (33) is valuable because it turns several apparently different perturbation models into one covariance projection problem. Three special cases are used in the experiment section: iid uniform box noise, common-bias motion, and correlated Gaussian errors. The model consequence is that only the component of the error covariance that survives the centered weighting $M_\alpha D_\xi$ can damage coherence significantly.

This covariance view is also what connects the deterministic and stochastic parts of the report. In the deterministic box problem, the harmful perturbation is the one that creates large centered phase spread within the allowed error budget. In the stochastic problem, the harmful law is the one whose covariance injects energy into the same centered differential directions. The two sections therefore use different mathematics, but they are driven by the same physical mechanism.

2.4 Numerical Setup, Baselines, and Reproducibility Contract

The experiments in this report are simulation based. No external dataset is used. Instead, the results are generated from deterministic and Monte Carlo evaluation of the PASS model in (4) under one fixed configuration. The report keeps the numerical workflow explicit because the scientific story depends on separating exact definitions from best-found numerical procedures.

Table 2 summarizes the exact configuration and notation used in the experiments.

Table 2: Core simulation configuration and notation used throughout the report.

Quantity	Value used in this report
Number of activated antennas N	16
Height d	3.0 m
Carrier frequency f_c	28 GHz
Wavelength λ	0.0107143 m
Effective refractive index n_{eff}	1.44
Minimum spacing factor Δ_p	0.5
ξ_{max}	1.4597
$\varepsilon_{\text{valid}}/\lambda$	0.1713
a_{ideal}	1.7775

The value $a_{\text{ideal}} = 1.7775$ in Table 2 is the design-referenced normalization constant for the perturbation study around Δ^* . It should not be confused with the absolute nominal gains reported later in the baseline section, where each placement is evaluated on a common raw-gain scale without the design-specific phase reference in (5).

This report concentrates on a single fully specified configuration, rather than distributing the main argument over several partially different settings. The choice is motivated by more than computational convenience: it keeps the evidence chain tight—all main-body figures and tables refer to a single constructive placement, a single a_{ideal} , a single validity threshold, and a single sensitivity profile. Once the behavior of that single configuration is clear, the auxiliary sweeps serve as mechanism checks, not as substitutes for the core argument.

All simulations use the configuration in Table 2 and examine the same constructive placement along two complementary tracks. The deterministic track handles bounded box errors by exact enumeration of corners and a continuous-box best-found refinement based on box-constrained local search. The stochastic track assesses i.i.d. uniform, common-bias, and correlated Gaussian perturbations, and compares their Monte Carlo averages with the second-order predictor in (33). The implementation employs a deterministic Python workflow: numerical evaluation is vectorized, random seeds are fixed for Monte Carlo sampling, and configuration snapshots are saved for each reported run. Together, the two tracks account for both small-error validity behavior and broader empirical trends.

The numerical campaign is organized as a sequence of reusable passes, rather than as a single opaque batch run. The constructive placement is generated once and stored as the reference

geometry; it is then reused unchanged in the bounded-error, stochastic, baseline, and sensitivity studies. This design is methodologically important: when two figures differ, the difference can be attributed to the perturbation model or the comparison task, rather than to an unannounced change in the nominal placement. In a long-form project report, such stability is more valuable than maximizing the number of loosely connected experiments.

Table 3 summarizes the numerical workflow at a level of detail appropriate to the manuscript. The table is included because a reader often needs to understand not only which equations were derived, but also how the evidence was produced—which sweeps are dense, which checkpoints receive extra validation, and which calculations are deterministic rather than Monte Carlo.

Table 3: Numerical workflow used to generate the report’s main evidence chain.

Task	Numerical budget	Role in the manuscript
Constructive reference design	One feasible wavelength-snapped placement for the fixed $N = 16$ geometry	Provides the common nominal geometry used by all main-body figures and tables.
Main bounded-error curve	41 error levels over $0 \leq \varepsilon/\lambda \leq 0.20$ with exact corner search and continuous-box refinement when needed	Supports the main robustness figure and the comparison with the deterministic lower bound.
Bounded-noise reference statistics	1000 Monte Carlo samples for each error level in the main bounded-noise curve	Shows how a typical random bounded perturbation compares with deliberately adversarial motion.
Box-validation checkpoints	5 dedicated levels $\{0.02, 0.05, 0.10, 0.15, 0.175\}$	Justifies the use of the term <i>box-best-found</i> and highlights the near-null behavior separately from the dense sweep.
Stochastic scenario validation	21 error levels over $0 \leq \varepsilon/\lambda \leq 0.10$ and 2000 samples for each of the three stochastic laws	Tests the covariance-based predictor against iid, common-bias, and correlated perturbations.
Same-aperture baselines	31 error levels over $0 \leq \varepsilon/\lambda \leq 0.15$ and four nominal-search restarts for the best-found comparator	Isolates the comparative design question under matched span and spacing constraints.
Secondary sensitivity sweeps	Fixed-absolute-error frequency sweep and three effective-index settings	Checks whether the same mechanism variables k_0 and n_{eff} also govern the observed sensitivity trends.

From a programming-method point of view, the implementation emphasizes vectorized array-gain evaluation, explicit separation between deterministic search and Monte Carlo sampling, and

fixed-seed reproducibility. In the main text, the simulation is presented as a numerical workflow, not a code listing. Consistent with the reporting guidance, the description foregrounds the computational logic, the sampling budgets, and the solver status, rather than a line-by-line account of the source. Detailed file-level information required for reproducibility is deferred to Appendix C.

The three stochastic models were chosen to span qualitatively distinct physical structures, not for completeness. Independent identically distributed uniform noise models bounded, uncorrelated placement offsets; common-bias motion captures a shared drift or calibration bias; correlated Gaussian perturbations generate smooth, coupled displacements across the aperture. Having all three allows one to examine not merely whether randomness is harmful, but which covariance structure is harmful. This distinction is necessary if the covariance predictor is to be read as a mechanistic explanation, not simply a fitted curve.

The baseline analysis is deliberately narrow. It does not aim at a universal PASS design problem. Instead, three placements that share the same aperture span are compared: the constructive aligned placement that anchors the main robustness argument; a naïve uniform-aperture placement formed by linearly spacing the positive-side coordinates between the fixed endpoints and mirroring them; and a best-found same-aperture nominal-gain comparator that searches interior coordinates under identical span and spacing constraints. Supplementary sweeps over carrier frequency and effective refractive index serve only as sensitivity checks, not as additional core claims.

The same-aperture restriction is critical. Without matched endpoints and a common spacing rule, a baseline table would confound aperture-size effects with placement-design effects, leaving the source of any performance difference opaque. By holding the span and the minimum spacing fixed, the comparison poses a narrower but more defensible question: what constructive alignment buys within essentially the same physical footprint. That is the level of comparative claim the present evidence can support.

2.4.1 Same-Aperture Baseline Construction

Let Δ_{pos}^* denote the positive-side coordinates of the constructive aligned placement, and let Δ_{min} and Δ_{max} be its first and last positive-side coordinates. The uniform-aperture baseline fixes the same endpoints and places the remaining positive-side coordinates uniformly between them, after which the negative side is completed by symmetry. The second comparator is a *best-found same-aperture nominal-gain baseline*, defined by

$$\max_{\Delta_{\text{pos}}} a(\text{mirror}(\Delta_{\text{pos}})) \quad (34)$$

subject to

$$\Delta_1 = \Delta_{\text{min}}, \quad \Delta_{N/2} = \Delta_{\text{max}}, \quad \Delta_{i+1} - \Delta_i \geq \Delta_p \lambda. \quad (35)$$

The optimization defined by (34)–(35) is solved via a multi-start L-BFGS-B procedure operating on the interior of the positive orthant, with four restarts [16]. The result thus represents the best nominal-gain figure identified under the same aperture and spacing constraints; it is not a

certified global optimum of the PASS problem. In contrast to Sections 3.1–3.2, the baseline section reports raw array-gain values so that different element placements can be compared on a common physical scale.

All numerical statements in the report originate from a single, consistent simulation campaign conducted under this protocol. The rounded values displayed in figures and tables are therefore drawn from the same set of stored outputs, ensuring that the wording of every claim can be traced to a common evidence base.

Table 4 maps the principal claims of the manuscript to the stored numerical artifacts. For a final-year report, such a mapping is instructive because it renders the distinctions among deterministic small-error assertions, empirical observations, and comparative claims explicit before the reader enters the results chapter.

Table 4: Claim-to-evidence map used to keep the manuscript wording aligned with the stored artifacts.

Claim or statement type	Primary artifact	Scope used in the manuscript
Tight deterministic lower bound in the first-order validity region	Stored configuration snapshot and main deterministic curve	Bound language restricted to $\varepsilon/\lambda \leq 0.1713$.
Pre-null split-mode near-adversarial behavior	Main deterministic curve and box-validation checkpoints	Empirical and restricted to the pre-null regime.
Second-order stochastic predictor explains iid, common-bias, and correlated scenarios	Stochastic scenario curve and Monte Carlo summary	Supported only for the studied design over $0 \leq \varepsilon/\lambda \leq 0.10$.
Same-aperture design comparison	Baseline summary table and baseline robustness curves	Narrow comparative statement across the three tested placements only.
Frequency and refractive-index sensitivity sweeps	Secondary sensitivity-sweep curves	Supporting evidence for mechanism interpretation, not a core generality claim.

The methods section therefore fixes the exact meaning of each later result. Deterministic lower-bound statements refer to the first-order validity region associated with (25). Empirical statements refer either to the pre-null box-best-found comparison or to the stochastic simulations. Comparative design claims are restricted to the same-aperture baseline section and do not imply broader design optimality.

3 Results

This section presents the numerical evidence in the same order as the claim logic established in Section 2. The goal is not to maximize the number of plots, but to present the minimum figure-and-table set needed to support the final conclusions with conservative wording.

3.1 Main Bounded-Error Robustness

The main deterministic robustness result is shown in Fig. 4. The plot compares the ideal normalized gain, the deterministic lower bound, the box-best-found adversary, the split-mode construction, the common-bias perturbation, and the Monte Carlo mean under iid box noise. Figure 3 serves as a mechanism cue for interpreting the plot: after centering the ξ_n profile, the sign pattern reverses at the array midpoint, making the split-mode perturbation a natural candidate for maximizing relative phase spread.

The figure is best read as a hierarchy of evidence, not as a set of comparable curves. The ideal line functions only as a normalization reference. The lower bound is the analytically justified object, though its validity is restricted to the first-order region marked on the plot. The box-best-found curve represents the strongest adversarial numerical estimate obtained in this report. The split-mode and common-bias curves act as mechanism probes: they test whether the centered-sensitivity interpretation correctly separates harmful from harmless directions within the same bounded uncertainty set. The Monte Carlo mean indicates how a typical random bounded perturbation compares with deliberately adversarial motion. Reading the plot in this order prevents the deterministic, empirical, and stochastic statements from merging into a single, over-strong claim.

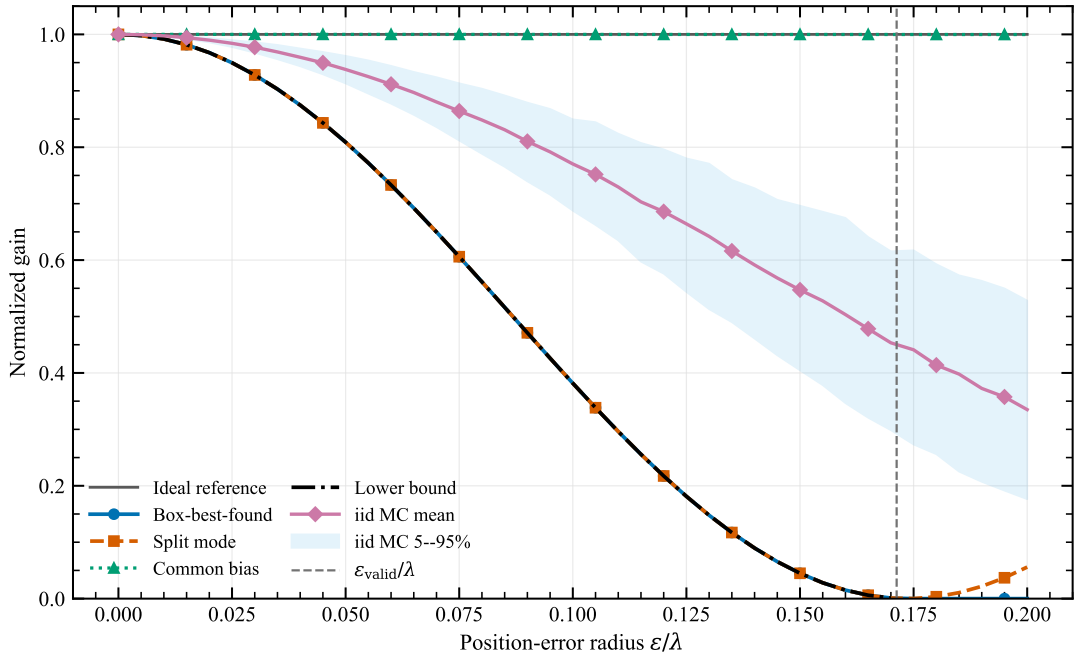


Figure 4: The main robustness curve for the constructive phase-aligned placement, depicted as design-referenced normalized gain. The vertical line indicates the first-order validity threshold, $\epsilon_{\text{valid}}/\lambda = 0.1713$. Within that region, the lower bound and the box-best-found curve are virtually indistinguishable.

Several observations stand out from the data. First, the deterministic lower bound is notably tight within the first-order validity region. The stored configuration yields $\varepsilon_{\text{valid}}/\lambda = 0.1712677618$, and over that interval the maximum absolute gap between the box-best-found curve and the lower bound is about 3.40×10^{-6} . This represents the strongest deterministic small-error evidence in the report. It implies that, for the chosen constructive design, the bounded-error degradation is essentially captured by the lower bound up to the reported validity threshold.

Second, the split-mode perturbation follows the box-best-found curve almost exactly in the pre-null regime. At $\varepsilon/\lambda = 0.05$, the box-best-found normalized gain is 0.808917 while the split-mode value is 0.808924; at $\varepsilon/\lambda = 0.10$ the two values are 0.381727 and 0.381743; at $\varepsilon/\lambda = 0.15$ they become 0.044941 and 0.044945. These close matches align with the centered-sensitivity argument of Section 2.3. Before the first null, the dominant harmful pattern is the left–right split, which enlarges the relative phase spread between the two halves of the array.

Third, the common-bias perturbation proves almost harmless across the whole tested range. Even at $\varepsilon/\lambda = 0.10$ the common-bias curve stays at 0.999939, and at $\varepsilon/\lambda = 0.15$ it remains 0.999862. A naive “all error is equally bad” narrative would overlook this behavior, but it follows directly from the phase-variance perspective: a nearly common phase rotation leaves the relative coherence of the weighted sum nearly intact.

Comparing the stochastic mean with the deterministic adversaries at the same error magnitude is also revealing. At $\varepsilon/\lambda = 0.10$, the iid Monte Carlo mean remains 0.769808, well above the box-best-found value of 0.381727. This gap quantifies how conservative the adversarial viewpoint is for the present design: most random perturbations do not align their signs to maximize the weighted phase spread. The bounded-box analysis is therefore valuable not as a predictor of typical behavior but as a tool for identifying the most harmful structured perturbations that a robust placement must survive.

Since the deterministic adversarial curves overlap visually over most of the range, their relationship is most clearly documented through discrete checkpoints. Table 5 reports the corner-restricted, split-mode, and box-best-found values at the five validation levels used in the dedicated box-validation pass.

Table 5: Main-body box-validation checkpoints for deterministic adversarial patterns, reported as design-referenced normalized gain values.

ε/λ	Corner-restricted	Split-mode	Box-best-found
0.02	0.9676082	0.9676089	0.9676082
0.05	0.8089171	0.8089240	0.8089171
0.10	0.3817270	0.3817426	0.3817270
0.15	0.0449408	0.0449452	0.0449408
0.175	1.5787×10^{-4}	1.5790×10^{-4}	1.88×10^{-17}

The first four rows show that, up to $\varepsilon/\lambda = 0.15$, exact corner enumeration already matches the continuous-box best-found value to displayed precision, while split mode differs only in the fifth or sixth decimal place. The last row is different. At $\varepsilon/\lambda = 0.175$, just beyond the first-order

validity region and near the first-null regime, the local refinement reaches numerical zero while the corner and split-mode candidates remain near 1.58×10^{-4} . This single row illustrates why precise wording is necessary. It confirms that the pre-null split-mode behavior is highly robust, yet it does not elevate the continuous search to a globally certified worst-case solver.

Appendix B presents the companion gap table, where the same checkpoints are reported as differences from the box-best-found value. This view is convenient for readers who wish to examine the small pre-null discrepancies without cluttering the main text with a nearly redundant table.

Section 3.1 thus supports two distinct statements, each with a different strength. Inside the first-order validity region, the lower bound essentially characterizes the box-best-found degradation of the chosen constructive design. Prior to the first null but outside the region of rigorous first-order justification, the split-mode curve provides an empirical, near-adversarial account of the observed degradation pattern. Maintaining this separation is essential to the claim discipline of the report.

3.2 Stochastic Scenario Validation

The stochastic scenario results are shown in Fig. 5. Each panel compares the simulated mean normalized gain with the second-order predictor from (33).

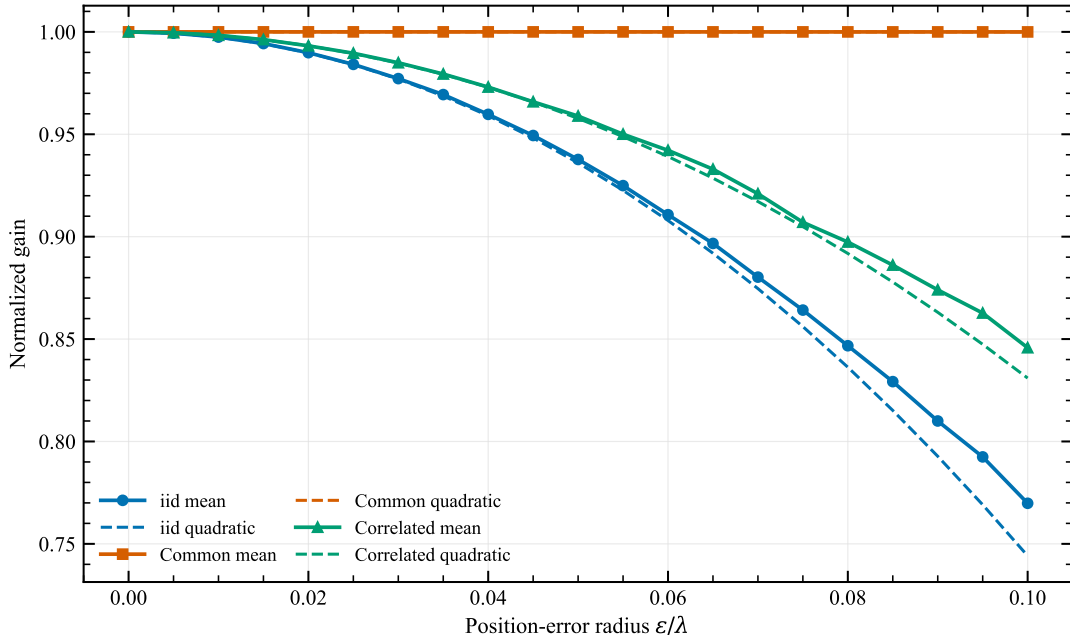


Figure 5: Stochastic error scenarios and quadratic phase-variance approximations, plotted as design-referenced normalized gain. The same second-order predictor explains iid box noise, common-bias motion, and correlated Gaussian perturbations over the tested range $0 \leq \varepsilon/\lambda \leq 0.10$.

At the representative point $\varepsilon/\lambda = 0.10$, the iid uniform mean is 0.769808, while the predictor gives 0.744161. The common-bias mean is 0.999979, and the predictor gives 0.999980. The correlated Gaussian mean is 0.845762, and the predictor gives 0.830947. These pointwise comparisons already show that the theory captures the qualitative ordering correctly: iid independent

perturbations are the most damaging among the tested stochastic laws, common-bias perturbations are nearly harmless, and correlated Gaussian errors lie between the two.

At the representative point $\varepsilon/\lambda = 0.10$, the iid uniform mean is 0.769808 and the predictor gives 0.744161; the common-bias mean is 0.999979 and the predictor gives 0.999980; the correlated Gaussian mean is 0.845762 and the predictor gives 0.830947. These pointwise comparisons already indicate that the theory correctly recovers the qualitative ordering: among the tested stochastic laws, iid independent perturbations are the most damaging, common-bias perturbations remain nearly harmless, and correlated Gaussian errors fall between the two.

The three panels retain their diagnostic value precisely because they do not collapse onto a single curve. If only the marginal error size mattered, then once the scenarios were matched in scale their mean degradations would appear far more similar. The observed ordering instead confirms that the covariance structure plays a decisive role. The iid model introduces differential phase errors across the entire aperture, the common-bias model is predominantly common-mode, and the correlated Gaussian model occupies an intermediate regime. The figure therefore lends comparable support to the covariance interpretation and to the second-order approximation.

The predictor also exhibits quantitative stability over the whole tested interval. For $0 \leq \varepsilon/\lambda \leq 0.10$, the mean absolute approximation error is approximately 5.67×10^{-3} for iid uniform noise, 8.41×10^{-8} for common bias, and 3.39×10^{-3} for correlated Gaussian errors. These error levels remain small enough to confirm that, although the weighted phase-variance predictor is not exact, it succeeds in capturing the stochastic trends observed across the studied covariance structures within the small-error range examined here.

The same quantitative behavior delineates where the approximation is reliable and where it should be treated with caution. The common-bias case is nearly exact because the centered phase variance removes almost all of its effect. The iid and correlated cases exhibit somewhat larger gaps, as higher-order phase terms and amplitude variations become non-negligible; nevertheless, the predictor preserves the correct scale and rank order. For the purposes of this report, that constitutes the appropriate level of success: the approximation functions as an explanatory tool supported by simulation, not as a substitute for it.

Table 6 condenses the approximation quality into error metrics computed across the full tested range. Placed in the body, this table complements the visual comparison in Fig. 5 by providing a compact experimental summary of how well the predictor maintains accuracy over the entire interval.

Table 6: Approximation-error summary for the stochastic predictor over the tested range $0 \leq \varepsilon/\lambda \leq 0.10$.

Scenario	Mean abs. error	Max abs. error	Max-error location
iid uniform	5.67×10^{-3}	2.56×10^{-2}	$\varepsilon/\lambda = 0.10$
Common bias	8.41×10^{-8}	3.66×10^{-7}	$\varepsilon/\lambda = 0.08$
Correlated Gaussian	3.39×10^{-3}	1.53×10^{-2}	$\varepsilon/\lambda = 0.095$

The table supports the same qualitative reading as the figure, but with two extra clarifications.

First, the common-bias case is not merely visually close to the predictor; it is numerically almost exact over the tested range. Second, the largest discrepancies for the iid and correlated cases occur near the upper end of the tested interval, which is exactly where one would expect higher-order phase effects to become more visible. This is further evidence that the second-order predictor is strongest as a small-error explanatory model rather than as a substitute for direct simulation at larger perturbation levels.

A smaller one-parameter experiment was added to isolate the covariance effect more directly. Instead of increasing the error radius, Fig. 6 fixes $\varepsilon/\lambda = 0.10$ and varies only the normalized correlation length of the Gaussian position-error field. The iid and common-bias references are held at the same error scale, so the curve can be read as a direct transition from differential error toward smoother, more common-mode error.

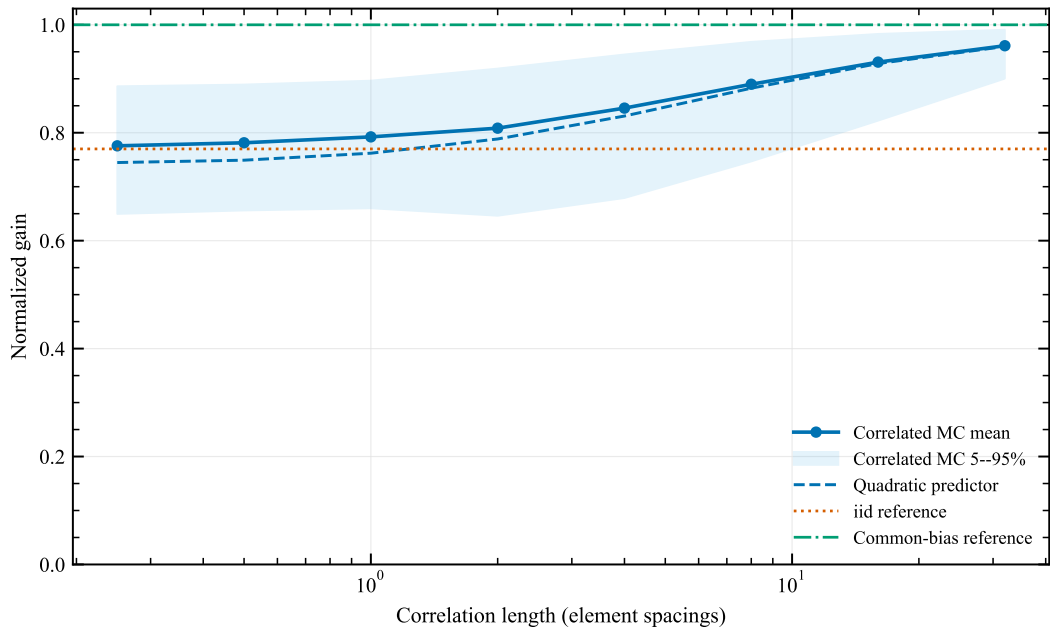


Figure 6: Correlation-length sensitivity at fixed $\varepsilon/\lambda = 0.10$ for correlated Gaussian position errors. The solid curve shows the Monte Carlo mean normalized gain, the shaded band shows the 5–95% interval, the dashed curve shows the quadratic predictor, and the horizontal references show iid uniform and common-bias means at the same error scale.

The sweep gives a more visual form of the stochastic mechanism. At normalized correlation length 0.25, the mean normalized gain is 0.7757, close to the iid reference value 0.7700. At correlation length 4, the mean is 0.8456, matching the correlated-Gaussian scenario used in Fig. 5. By correlation length 32, the mean rises to 0.9612, moving toward the common-bias reference value 0.99998. The quadratic predictor follows the same monotone trend, which supports the interpretation that smoother error fields cause less centered phase variance across the aperture. This is a sensitivity check for the studied design and error scale, not a general theorem about all PASS layouts.

Table 7: Representative checkpoints for the deterministic and stochastic robustness story.

Quantity	$\varepsilon/\lambda = 0.05$	$\varepsilon/\lambda = 0.10$	$\varepsilon/\lambda = 0.15$
Box-best-found normalized gain	0.8089	0.3817	0.04494
Split-mode normalized gain	0.8089	0.3817	0.04495
Common-bias normalized gain	0.99998	0.99994	0.99986
Lower-bound normalized gain	0.8089	0.3817	0.04494
iid mean normalized gain	0.9378	0.7698	—
Correlated-Gaussian mean normalized gain	0.9589	0.8458	—

3.3 Placement Baseline Comparison

The baseline section converts the report from a one-design robustness note into a narrower design-comparison story. Figure 7 compares the box-best-found curves of the three same-aperture placements, while Fig. 8 summarizes the nominal-versus-robustness tradeoff.

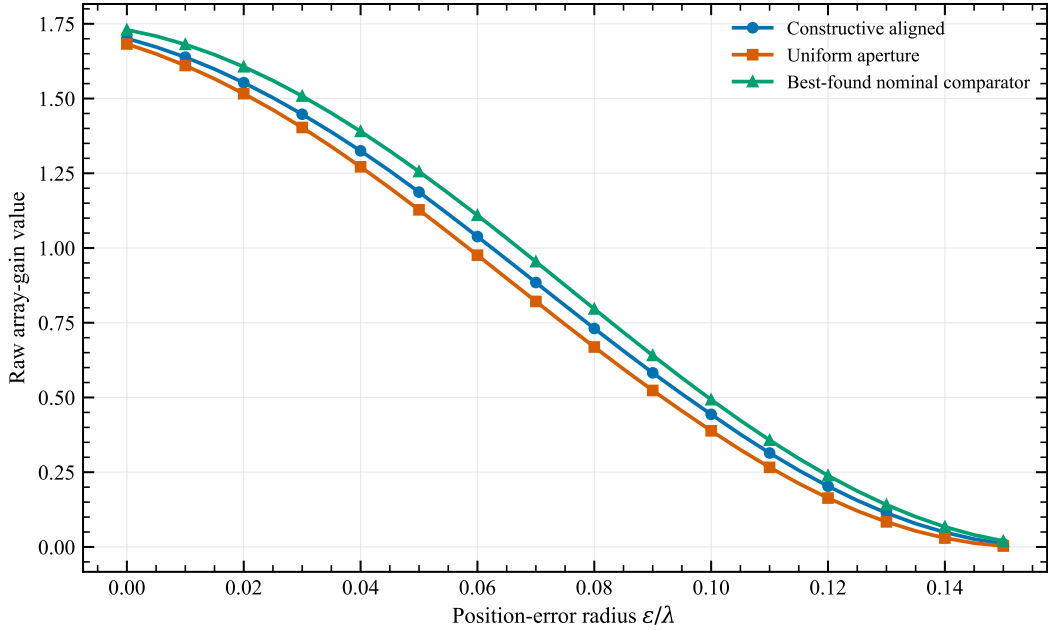


Figure 7: Same-aperture box-best-found robustness curves in raw array-gain value for the constructive aligned placement, a naive uniform-aperture baseline, and the best-found nominal-gain comparator.

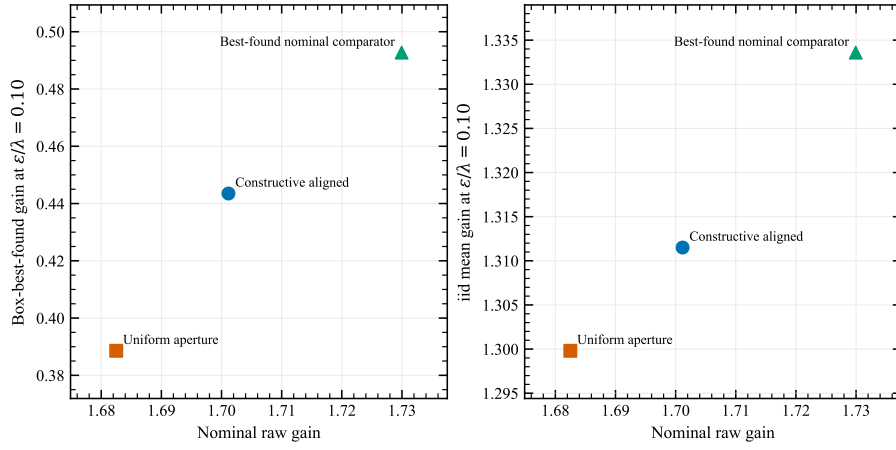


Figure 8: Nominal-gain versus robustness tradeoff across the three same-aperture placements at the reference error level, shown in raw array-gain value. The constructive aligned design is better than the naive uniform baseline, but the best-found nominal-gain comparator dominates both in the current comparison.

Table 8 summarizes the most important checkpoints.

Unlike the normalized perturbation curves in Sections 3.1–3.2, Table 8 reports *raw array-gain values*. This is deliberate: once different placements are compared, a common raw scale is more informative than normalizing each design by its own nominal ideal, because it preserves both nominal-performance differences and robustness differences in one table. In particular, the constructive aligned nominal value 1.7012 in Table 8 is the raw gain of that placement, whereas the earlier value $a_{\text{ideal}} = 1.7775$ is the alignment-based normalization constant used for the perturbation study around the same reference design.

Making that scale choice explicit matters for interpreting the constructive design fairly. If each placement were normalized by its own ideal reference, a design with weaker nominal coherent addition could still appear deceptively competitive after rescaling. The raw-gain presentation avoids that problem and keeps the baseline section anchored to one physically comparable scale. It also prevents the baseline discussion from sounding inconsistent with the earlier perturbation-focused sections, which necessarily use design-referenced normalization for a different purpose.

Table 8: Same-aperture baseline summary at $\varepsilon/\lambda = 0.10$ using raw array-gain values; the third column is a best-found nominal-gain comparator under the same aperture and spacing constraints.

Quantity	Uniform aperture	Constructive aligned	Best-found nominal comparator
Nominal gain	1.6825	1.7012	1.7299
Box-best-found gain	0.3886	0.4435	0.4927
iid mean gain	1.2998	1.3115	1.3336
Common-bias mean gain	1.6825	1.7011	1.7299
Correlated-Gaussian mean gain	1.4312	1.4430	1.4651

The constructive aligned design improves upon the naive uniform-aperture baseline, but the gain is not uniform across metrics. Relative to the uniform baseline, the constructive design improves nominal gain by about 1.11%, improves iid mean robustness by about 0.90%, and improves

box-best-found robustness by about 14.14%. This is a meaningful, but narrow, result. It shows that the phase-aligned constructive rule is especially effective against the damaging phase-spread behavior identified by the box-best-found search in the studied aperture-matched setting.

That unequal improvement pattern is itself informative. The constructive rule does not transform the nominal coherent sum dramatically, but it changes the geometry enough to make the most harmful bounded perturbations much less effective than they are for the uniform baseline. In other words, its value is not only that it aligns phases at the operating point, but that it appears to produce a sensitivity profile whose centered spread grows more slowly under adversarial box-feasible motion. This is still an empirical interpretation, yet it is exactly the type of design insight that a longer report should extract from a baseline table rather than leaving implicit.

At the same time, the best-found nominal-gain comparator performs better than the constructive aligned placement on every reported metric in this comparison. It improves nominal gain by about 1.69%, box-best-found robustness by about 11.09%, and iid mean robustness by about 1.68% relative to the constructive design. The report therefore does *not* conclude that the constructive aligned placement is the best same-aperture design. The correct conclusion is more modest: the constructive aligned rule has a clear advantage over a naive uniform placement, but a stronger nominal-gain baseline can do better in the current tested setup. Since this comparator is still not a robustness-tuned design, the ranking should be read as an empirical result for the tested placements rather than as evidence that nominal-gain maximization always improves robustness.

3.4 Secondary Sensitivity Sweeps

The secondary sweeps reinforce the mechanism interpretation without changing the main claim story. In the frequency sweep, a fixed absolute error of 1 mm yields a box-best-found normalized gain of approximately 0.9676 at 6 GHz, 0.4411 at 28 GHz, and numerical zero at 60 GHz. This follows the factor k_0 in both the deterministic lower bound and the phase-variance predictor: the same physical placement error becomes much more damaging as carrier frequency rises. The full frequency curve is kept in Appendix B because it mainly confirms a familiar scaling law once the phase model is fixed.

The effective-index sweep is more PASS specific and is therefore promoted to the main body in Fig. 9. Unlike carrier frequency, which rescales the phase error through k_0 , the effective refractive index changes the guided-wave contribution that distinguishes PASS from a purely free-space movable-antenna model in the first place.

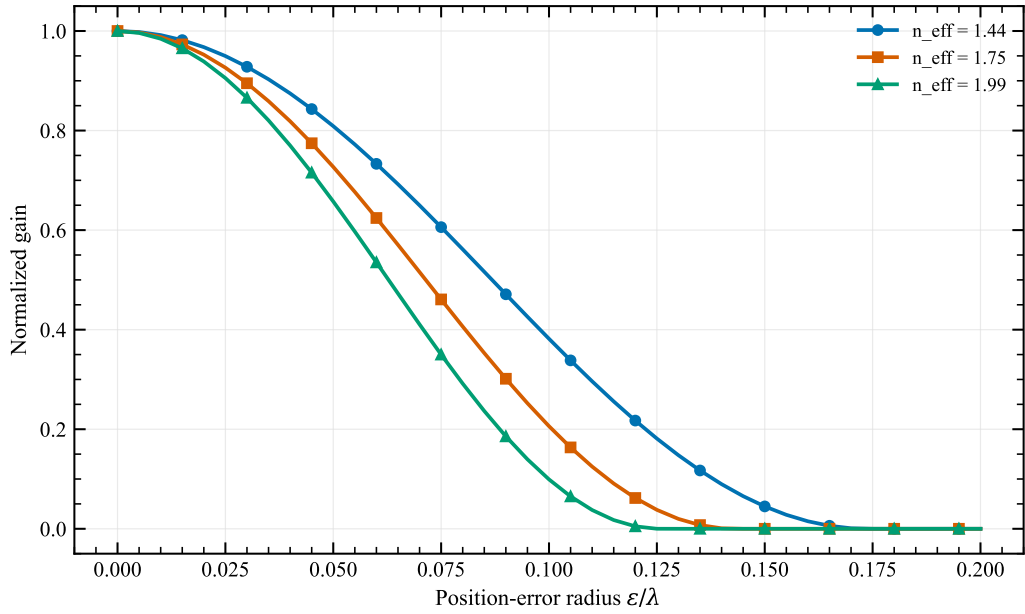


Figure 9: Sweep over the effective refractive index n_{eff} , plotted as design-referenced normalized box-best-found gain. Larger guided-wave phase contribution shifts the sensitivity profile upward and makes the same normalized position error more damaging.

In the n_{eff} sweep, the box-best-found normalized gain at $\epsilon/\lambda = 0.10$ drops from 0.3817 when $n_{\text{eff}} = 1.44$ to 0.2061 when $n_{\text{eff}} = 1.75$, and further to 0.0992 when $n_{\text{eff}} = 1.99$. This trend is consistent with the sensitivity model, because the whole ξ_n profile is shifted upward by the effective refractive index. Figure 9 therefore plays a different role from the baseline plots in Section 3.3. It is not a design comparison across placements. Instead, it is a mechanism check showing that the same PASS-specific coefficient emphasized in the derivation also controls the observed deterioration trend when the waveguide phase contribution increases.

Table 9 provides a compact checkpoint summary for the two sensitivity sweeps. The two blocks should not be compared row by row, because the frequency sweep fixes an absolute error of 1 mm while the effective-index sweep fixes a normalized error level $\epsilon/\lambda = 0.10$. Their purpose is simply to make the sweep magnitudes visible in one place without forcing the reader to extract individual values from separate curves.

Table 9: Checkpoint summary for the secondary sensitivity sweeps.

Sweep	Setting	Fixed error level	Box-best-found norm. gain
Carrier frequency	6 GHz	1 mm absolute error	0.9676
Carrier frequency	28 GHz	1 mm absolute error	0.4411
Carrier frequency	60 GHz	1 mm absolute error	Numerical zero
Effective refractive index	$n_{\text{eff}} = 1.44$	$\epsilon/\lambda = 0.10$	0.3817
Effective refractive index	$n_{\text{eff}} = 1.75$	$\epsilon/\lambda = 0.10$	0.2061
Effective refractive index	$n_{\text{eff}} = 1.99$	$\epsilon/\lambda = 0.10$	0.0992

The table below aids the interpretation of Section 3.4. For a fixed absolute placement error,

raising the frequency from 6 GHz to 28 GHz cuts the normalized worst-case gain by more than half; at 60 GHz the gain effectively vanishes. Under a fixed normalized error level, increasing n_{eff} from 1.44 to 1.99 reduces the same quantity from 0.3817 to 0.0992. These observations do not constitute new theorems, but they solidify the experimental evidence that both k_0 and n_{eff} act as multiplicative stress factors for robustness in the studied PASS configuration.

The purpose of these sweeps is therefore interpretive, not expansionary: they confirm that the same coefficients central to the theory— k_0 and n_{eff} —also drive the sensitivity trends when the configuration is perturbed. While this agreement yields no additional theorem, it reinforces the view that the reported mechanisms are rooted in the PASS phase law, rather than in an accidental numerical instance. In the structure of the report, Section 3.4 stress-tests the main robustness argument against physically meaningful parameter variations without extending the claims beyond the geometry under investigation.

4 Discussion

The results in Section 3 support a coherent story, but the value of a graduation report lies not only in showing curves and tables. It lies in drawing the right conclusions from them and making the scope of those conclusions explicit. This section therefore turns the numerical evidence into a disciplined interpretation.

4.1 Mechanism Interpretation

Weighted phase variance is central to this project. After the nominal constructive design is normalized by its amplitude weights, the coherent sum depends primarily on the spread of elementwise phase errors about their weighted mean. This framework accounts for all major observations more economically than the earlier same-sign perturbation narrative.

We first clarify why common-bias motion is nearly harmless. When all activated pinching antennas shift in roughly the same effective phase direction, the perturbation contributes mainly to a common phase term rather than to a relative spread. In the second-order expansion that common term enters the weighted mean and cancels from the variance. The common-bias curve remains almost exactly at unity, which corroborates the phase-variance interpretation; it is not an isolated numerical accident.

The same mechanism also explains the striking efficacy of the split-mode perturbation in the present symmetric configuration. The constructive placement renders the positive- and negative-side geometries nearly mirror-symmetric, and by symmetry the weighted mean of the sensitivity profile stays precisely at n_{eff} . Once $\xi_n - \bar{\xi}_\alpha$ is centered about zero, the error pattern that maximizes the local phase spread naturally drives the right and left halves in opposite directions. The resulting adversary is conceptually simple yet physically meaningful: it achieves the largest relative phase separation across the aperture while respecting the error budget. The close agreement between the split mode and the box-best-found adversary before the first null confirms that this local mechanism is not merely descriptive—it is predictive in the pre-null regime.

The stochastic results provide further support. With independent perturbations the covariance energy sits on the diagonal, and the centered weighting in (33) therefore yields a large elementwise phase spread. Common-bias motion remains nearly invisible because its covariance has rank one in the all-ones direction and is largely annihilated by the centered operator. Spatially correlated Gaussian perturbations fall between these extremes, as their smooth structure induces partial common-mode and partial differential behavior. Thus the stochastic section is not a supplementary addendum; it constitutes a second, independent validation of the very same weighted phase-variance object that already rationalizes the deterministic split-mode behavior.

4.2 Validity Condition Versus Empirical Region

A central refinement introduced during this project is the explicit separation of consequences that follow from the deterministic small-error validity condition and those that are merely empirical. The report preserves this distinction for both the lower bound and the worst-case search.

Inside the small-error region defined by $k_0 \xi_{\max} \varepsilon \leq \pi/2$, the projection step employed in the lower-bound derivation holds under the first-order phase-deviation model. Accordingly, the recorded threshold $\varepsilon/\lambda = 0.1713$ is not a fitted parameter; it is the direct output of the analytic condition on which the bound rests. That the lower bound coincides with the box-best-found curve to within approximately 3.40×10^{-6} over the entire interval constitutes strong evidence of tightness for the studied design. In this sense, the report’s strongest claim is neither the split-mode account nor the baseline comparison, but the tight deterministic small-error characterization valid within the stated region.

Beyond that threshold, the report deliberately shifts its terminology. The lower bound may remain close to the observed curve for some range, yet such agreement is empirical. Similarly, the continuous-box local-search routine can return values lower than the corner-restricted adversary, but this does not certify a global continuous minimum. For this reason the term *box-best-found* is used consistently. The language is conservative by necessity: the nonconvex bounded-box problem is not globally solved here.

The near-null point around $\varepsilon/\lambda = 0.175$ offers the cleanest illustration of why this distinction matters. Up to $\varepsilon/\lambda = 0.15$, the corner solution, the split-mode construction, and the box-best-found local search all agree to high precision. A cursory reading might then conclude that the corners are always exact or that the split-mode solution is exact. The 0.175 case shows why such a conclusion would be premature. There the local search reaches numerical zero while the corner and split-mode candidates remain at about 1.58×10^{-4} . The correct interpretation is that the pre-null local argument is very strong, but not globally universal. The report therefore treats the pre-null split-mode pattern as an empirical near-adversarial construction, not as a theorem.

4.3 Comparative Design Takeaway

The baseline section adds value, but only if its meaning is kept narrow. The constructive aligned placement should not be advertised as the best design in the same-aperture comparison, because it is not. The best-found nominal-gain comparator is stronger on nominal gain, on box-best-found robustness, and on iid mean robustness. This is a concrete result, not a minor caveat.

At the same time, the constructive aligned design clearly does outperform the naive uniform-aperture baseline, especially under box-best-found robustness. The pattern of the gains is informative. Improvements in nominal gain and iid mean are both around one percent, whereas the improvement in box-best-found robustness exceeds fourteen percent. This difference suggests that the constructive alignment rule is particularly good at controlling the harmful phase-spread direction identified by the split-mode analysis. In essence, the constructive design offers an improvement over the nominal placement that is more than marginal. Its phase sensitivity distribution appears to mitigate the box-best-found degradation relative to that observed in a uniformly spaced aperture of identical span.

For the purposes of the present report, the appropriate design-level conclusion can be stated as follows: the constructive phase-aligned feasible placement yields a meaningful gain over a naive same-aperture uniform baseline in the studied robustness problem, while falling short of the best

placement tested overall. This claim is scientifically valuable precisely because its scope is limited. It communicates that the phase-aligned constructive design makes a non-vacuous contribution, and it simultaneously confirms that scope remains for further gains through robustness-aware or nominally optimized placement strategies.

4.4 Engineering Implications of the Sensitivity Checks

The secondary sweeps presented in Section 3.4 derive their value from translating the underlying local mechanism into engineering terms. The effective-index sweep is particularly instructive. Raising n_{eff} does more than rescale a coordinate axis; it amplifies the guided-wave contribution within the PASS phase law itself. The associated decrease in the box-best-found normalized gain accordingly strengthens the interpretation that PASS robustness arises not solely from free-space geometry but from the joint geometry–waveguide phase accumulation that produces constructive alignment.

The frequency sweep offers a complementary insight. When a fixed absolute placement error is specified in meters rather than wavelengths, the effective perturbation grows more pronounced as the carrier frequency increases. Thus, a placement-accuracy budget that is adequate at 6 GHz may become unacceptable at 28 GHz or 60 GHz, even when the same mechanical actuation hardware is employed. For PASS systems operating at higher frequencies, preserving an equivalent robustness margin would therefore require tighter position control, shorter effective apertures, or more extensive active calibration.

Taken together, the common-bias result and the sensitivity sweeps suggest a clear priority for future control strategies. Given that common drift is largely benign whereas differential motion is destructive, the most valuable correction effort may not be whole-aperture translation compensation. Instead, suppressing or calibrating the differential error modes that widen the centered phase spread would prove more effective. While this falls short of a closed-loop demonstration, it represents a concrete design insight derived from the current evidence.

These implications are subject to the same scope constraints as the remainder of the report. They constitute design-specific interpretations for the symmetric configuration examined, not universal PASS scaling laws. Their contribution lies in demonstrating that the main theoretical coefficients are more than abstract symbols—they also identify the conditions under which robustness pressure would be expected to intensify in future engineering realizations of the problem.

4.5 Limitations and Unsupported Claims

Several limitations warrant attention. First, the bounded-box continuous search yields only best-found solutions, not provably global optima. Although integrating exact corner enumeration with box-constrained local refinement markedly outperforms a corner-only strategy, the overall procedure remains a heuristic for the continuous formulation. This restriction is most consequential near the first null, where the objective landscape becomes sharply nonconvex.

Second, the split-mode construction is an empirically identified near-adversarial pattern rather

than an exact minimizer statement. The weighted phase-variance derivation explains why the construction appears compelling, yet the reasoning is fundamentally local and confined to small-error regimes. Global optimality over the full box for arbitrary error magnitudes is not established.

Third, the stochastic predictor has been validated solely for the studied constructive design and only up to $\varepsilon/\lambda = 0.10$. Within that interval the approximation error remains modest, but the theory does not claim to retain comparable accuracy under larger errors, substantially different placements, or arbitrary covariance structures.

Fourth, the comparative design evaluation is limited in scope. The baseline analysis does not sweep over N , d , or Δ_p , and it includes no explicitly robustness-tuned alternative placement. Consequently, the observation that the constructive aligned placement outperforms a naive uniform baseline while underperforming the best-found nominal-gain comparator is a context-dependent finding rather than a universal design ranking.

Taken together, these constraints delineate the report’s scientific standing. The contribution is a meticulous robustness characterization of one constructive PASS design within a single symmetric configuration, grounded in reproducible simulations and a coherent local theory. It does not constitute a universal theorem of PASS robustness, nor does it survey the full PASS design space. The report’s strength is that it makes these boundaries explicit, rather than concealing them behind stronger rhetoric.

5 Conclusions

This report set out to study the robustness of a constructive phase-aligned PASS placement under position uncertainty while preserving claim discipline. Starting from the array-gain model of [10], it reformulated the nominal design as a feasible wavelength-snapped phase-alignment construction, derived a position-dependent phase sensitivity, established a deterministic lower bound in the small-error regime, and used a weighted phase-variance expansion to explain both deterministic and stochastic degradation behavior.

5.1 Conclusions

Four conclusions are supported by the current evidence.

First, for the constructive phase-aligned placement studied here, the deterministic lower bound provides an exceptionally tight characterization of bounded-error degradation inside the first-order small-error validity region. The threshold $\varepsilon/\lambda = 0.1713$ is analytically derived from the phase-sensitivity bound, and the maximum gap between the lower bound and the box-best-found curve over that region is only about 3.40×10^{-6} . This is the strongest and most reliable technical result in the report.

Second, in the studied symmetric setting, the dominant harmful bounded-error pattern before the first null is not a common signed shift. It is an empirical left-right split-mode phase-spread pattern. The split-mode construction derived from the centered-sensitivity profile tracks the box-best-found curve almost perfectly up to $\varepsilon/\lambda = 0.15$, while the common-bias perturbation keeps the normalized gain essentially at one over the same range. This gives both an interpretable physical mechanism and a practically useful adversarial surrogate.

Third, the same weighted phase-variance object explains the stochastic scenario study over the tested range $0 \leq \varepsilon/\lambda \leq 0.10$. The second-order covariance-based predictor captures the observed behavior of iid uniform, common-bias, and correlated Gaussian perturbations for the studied constructive design. The predictor does not claim exactness, but it successfully explains why independent perturbations are damaging, common shifts are nearly harmless, and correlated perturbations produce intermediate degradation.

Fourth, the baseline section supports only a narrow comparative design claim. In the tested same-aperture comparison, the constructive aligned placement is more robust than a naive uniform-aperture baseline, with the largest gain appearing in box-best-found robustness. However, the best-found nominal-gain comparator performs better than the constructive aligned design on every reported metric in that comparison. The report therefore does not recommend the constructive placement as the best overall design. It only shows that the constructive rule is materially better than a naive uniform placement in the current setup.

The broader methodological conclusion is equally important. A robustness study in this setting becomes much clearer once one keeps deterministic small-error results, best-found numerical results, empirical mechanism interpretations, and comparative design claims separate. The weighted phase-variance framework is useful precisely because it supports that separation. It

yields a lower-bound interpretation in the first-order validity region, an empirical split-mode interpretation before the first null, and a covariance projection for stochastic perturbations, all without forcing the report to overstate what is actually proved.

Considered as an integrated project-report argument, the work amounts to considerably more than a compilation of plots. A single constructive placement strategy is maintained throughout—from model definition and design construction, across deterministic and stochastic perturbation analyses, and into a narrow same-aperture comparison. This continuity matters because it allows the report to delineate precisely which elements are analytical, which are numerical, and which are merely comparative. The outcome is not a universal PASS design theorem but a carefully bounded robustness case study with well-defined limits.

The practical interpretation is relatively straightforward. For the design under examination, the principal challenge lies not in common motion of the full aperture but in differential motion that broadens the centered phase spread. Sensitivity intensifies when the waveguide contribution grows in relative importance or when identical absolute mechanical errors register at higher carrier frequencies. These are the kinds of engineering demands that future PASS implementations would face if constructive alignment is to retain its utility beyond idealized nominal placements.

5.2 Suggestions for Further Work

The next phase of this work should broaden the investigation rather than merely add curves of the same class. Four lines of inquiry appear particularly promising.

One direction is to intensify the continuous bounded-box search in the vicinity of the first null. Introducing additional random restarts, employing alternative local solvers, or performing an independent global-search cross-check would help ascertain whether the near-null gap observed between corner patterns and the box-best-found outcome reflects a genuine structural property or a numerical artifact of the present routine.

A second extension would broaden the baseline comparison. Rather than introducing another minor variant of the constructive placement, a more informative next step is a placement-family study that systematically varies antenna count and spacing, and includes at least one design explicitly tuned for robustness. This shifts the focus from a single local design conclusion to a more general investigation of placement strategies.

Third, the stochastic framework warrants testing on a broader set of geometries and covariance structures beyond those considered here. Although the current covariance-based predictor performs adequately for the studied design over a modest error range, it remains to be determined which aspects of the predictor remain stable when the aperture span changes, when n_{eff} is varied substantially, or when the perturbation field has stronger long-range correlation.

Fourth, the work could be extended toward a more comprehensive PASS robustness framework by incorporating actuation or calibration models. At present the perturbation model treats error fields as exogenous; a more engineering-oriented formulation would link those errors to actual pinching hardware, calibration expense, and potentially to closed-loop correction schemes.

In summary, this report provides a credible and reproducible foundation for PASS robustness

analysis. Its contribution lies not in resolving all open questions, but in bringing the project to a point where the next technical steps are clearly defined and well motivated.

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A Detailed Derivations

This appendix records the detailed derivation chain behind the main text. The goal is not to introduce stronger claims than those used in the body. It is to show clearly how the formulas used in the report arise from the PASS model, and which steps are exact identities, which are local approximations, and which are interpretation-driven constructive choices.

A.1 Derivation of the Constructive Phase-Alignment Equation

Start from the inherited array-gain expression

$$a(\Delta) = \frac{\eta}{N} \left| \sum_n \frac{e^{-jk_0(R_n(\Delta_n) + n_{\text{eff}}\Delta_n)}}{R_n(\Delta_n)} \right|^2, \quad R_n(\Delta_n) = \sqrt{d^2 + \Delta_n^2}. \quad (36)$$

The coherent-sum envelope satisfies

$$\left| \sum_n c_n e^{-j\theta_n} \right| \leq \sum_n |c_n|, \quad (37)$$

with equality when all phases are equal modulo 2π . In the present problem $c_n = R_n(\Delta_n)^{-1}$ is real and positive, so a constructive alignment target is

$$k_0(R_n(\Delta_n) + n_{\text{eff}}\Delta_n) = C + 2\pi m_n, \quad m_n \in \mathbb{Z}. \quad (38)$$

Dividing by $k_0 = 2\pi/\lambda$ gives

$$R_n(\Delta_n) + n_{\text{eff}}\Delta_n = C' + m_n\lambda, \quad C' = \frac{C}{k_0}. \quad (39)$$

The report does not optimize jointly over all integers m_n . Instead, it uses a constructive feasible scheme in which the wavelength target on the positive side is snapped upward from an initialization grid:

$$\Delta_n^{(0)} = \frac{\Delta_p\lambda}{2} + (n-1)\Delta_p\lambda, \quad n = 1, \dots, \frac{N}{2}. \quad (40)$$

The snapped target is

$$d_n = \lambda \left\lceil \frac{\sqrt{d^2 + (\Delta_n^{(0)})^2} + n_{\text{eff}}\Delta_n^{(0)}}{\lambda} \right\rceil, \quad (41)$$

and the aligned position then solves

$$\sqrt{d^2 + \Delta_n^2} + n_{\text{eff}}\Delta_n - d_n = 0. \quad (42)$$

A.2 Closed-Form Solution of the Scalar Alignment Equation

Equation (42) can be solved numerically, but it is useful to see its closed form because it reveals the admissible branch structure. Rearranging gives

$$\sqrt{d^2 + \Delta_n^2} = d_n - n_{\text{eff}}\Delta_n. \quad (43)$$

Squaring both sides yields

$$d^2 + \Delta_n^2 = d_n^2 - 2d_n n_{\text{eff}}\Delta_n + n_{\text{eff}}^2 \Delta_n^2, \quad (44)$$

and therefore

$$(n_{\text{eff}}^2 - 1)\Delta_n^2 - 2d_n n_{\text{eff}}\Delta_n + (d_n^2 - d^2) = 0. \quad (45)$$

For $n_{\text{eff}} \neq 1$, the two algebraic roots are

$$\Delta_n = \frac{d_n n_{\text{eff}} \pm \sqrt{d_n^2 + d^2(n_{\text{eff}}^2 - 1)}}{n_{\text{eff}}^2 - 1}. \quad (46)$$

The positive-side constructive placement uses the smaller root, because the larger root corresponds to a much farther coordinate on the same phase branch. In practice the code uses a bracketed scalar root finder and then verifies the spacing constraint with the previously accepted position. If the spacing is too small, d_n is increased by one more wavelength and the solve is repeated. This means that the constructive algorithm should be understood as a feasible snapping-and-repair procedure, not as a global nominal optimizer.

A.3 Derivation of the Phase Sensitivity

The phase of the n th term in (36) is

$$\Phi_n(\Delta_n) = k_0(R_n(\Delta_n) + n_{\text{eff}}\Delta_n). \quad (47)$$

Differentiating with respect to Δ_n gives

$$\Phi'_n(\Delta_n) = k_0 \left(\frac{\partial R_n(\Delta_n)}{\partial \Delta_n} + n_{\text{eff}} \right) = k_0 \left(\frac{\Delta_n}{\sqrt{d^2 + \Delta_n^2}} + n_{\text{eff}} \right). \quad (48)$$

Evaluated at the constructive placement,

$$\xi_n \triangleq \frac{1}{k_0} \Phi'_n(\Delta_n^*) = \frac{\Delta_n^*}{\sqrt{d^2 + (\Delta_n^*)^2}} + n_{\text{eff}} = \sin \theta_n^* + n_{\text{eff}}. \quad (49)$$

If $\Delta_n = \Delta_n^* + \delta_n$ and $|\delta_n|$ is small, a first-order Taylor expansion gives

$$\Phi_n(\Delta_n^* + \delta_n) \approx \Phi_n(\Delta_n^*) + k_0 \xi_n \delta_n, \quad (50)$$

hence the leading phase error is

$$\Delta\phi_n \approx k_0 \xi_n \delta_n. \quad (51)$$

A.4 Deterministic Lower Bound Under the First-Order Phase Condition

Write

$$w_n^* = \frac{1}{R_n(\Delta_n^*)}, \quad w_n^- = \frac{1}{\sqrt{d^2 + (|\Delta_n^*| + \varepsilon)^2}}. \quad (52)$$

Since $|\delta_n| \leq \varepsilon$, the perturbed amplitude satisfies

$$\frac{1}{R_n(\Delta_n^* + \delta_n)} \geq w_n^-. \quad (53)$$

Using the first-order phase law and the worst-case projection bound,

$$\Re \left(e^{-j\Delta\phi_n} \right) \geq \cos(k_0 |\xi_n| \varepsilon) \quad \text{whenever} \quad k_0 |\xi_n| \varepsilon \leq \frac{\pi}{2}. \quad (54)$$

Now define the perturbed coherent sum

$$S(\delta) = \sum_n \frac{e^{-j(\Phi_n(\Delta_n^* + \delta_n) - \Phi_n(\Delta_n^*))}}{R_n(\Delta_n^* + \delta_n)}. \quad (55)$$

Because $|S(\delta)| \geq \Re S(\delta)$, we have

$$|S(\delta)| \geq \sum_n w_n^- \cos(k_0 |\xi_n| \varepsilon). \quad (56)$$

Dividing by the ideal aligned reference $\sum_n w_n^*$ and squaring gives the normalized lower bound for the phase-referenced diagnostic used in the main text:

$$\tilde{a}(\Delta^* + \delta; \Delta^*) \geq \left(\frac{\sum_n w_n^- \cos(k_0 |\xi_n| \varepsilon)}{\sum_m w_m^*} \right)^2. \quad (57)$$

A.5 Derivation of the Weighted Phase-Variance Expansion

The phase-variance interpretation begins by normalizing the amplitudes:

$$\alpha_n = \frac{w_n^*}{\sum_m w_m^*}, \quad \sum_n \alpha_n = 1. \quad (58)$$

Ignoring higher-order amplitude changes for mechanism interpretation, the normalized coherent sum is approximated by

$$\rho(\delta) = \sum_n \alpha_n e^{-jk_0 \xi_n \delta_n}. \quad (59)$$

For small phase errors,

$$e^{-jk_0\xi_n\delta_n} = 1 - jk_0\xi_n\delta_n - \frac{1}{2}k_0^2\xi_n^2\delta_n^2 + O(\|\delta\|^3). \quad (60)$$

Substituting into (59) gives

$$\rho(\delta) = 1 - j \sum_n \alpha_n k_0 \xi_n \delta_n - \frac{1}{2} \sum_n \alpha_n (k_0 \xi_n \delta_n)^2 + O(\|\delta\|^3). \quad (61)$$

Let $\mu_\phi = \sum_n \alpha_n k_0 \xi_n \delta_n$. Then, up to second order,

$$|\rho(\delta)|^2 \approx 1 - \sum_n \alpha_n (k_0 \xi_n \delta_n)^2 + \mu_\phi^2 = 1 - \text{Var}_\alpha(k_0 \xi_n \delta_n). \quad (62)$$

This is the local object that unifies the deterministic and stochastic interpretations in the main text.

B Additional Results and Sensitivity Sweeps

This appendix collects the numerical tables and secondary curves that support the main text but would interrupt the flow if kept in the core body.

B.1 Box Validation Gaps

Section 3.1 now carries the value table needed for the main argument. This appendix retains the companion gap table, which makes the tiny pre-null discrepancies easier to inspect without adding a second nearly redundant checkpoint table to the body.

Table 10: Box-validation checkpoints for deterministic adversarial patterns, reported as gaps in design-referenced normalized gain relative to the box-best-found value.

ε/λ	Corner minus box	Split minus box
0.02	0	6.69×10^{-7}
0.05	0	6.91×10^{-6}
0.10	0	1.57×10^{-5}
0.15	0	4.42×10^{-6}
0.175	1.5787×10^{-4}	1.5790×10^{-4}

This table makes the report’s wording choice transparent. Up to 0.15, the continuous-box refinement does not improve on the best corner to displayed precision. At 0.175, it does. This is why the terminology must change from corner-restricted to box-best-found once the continuous search is introduced.

B.2 Frequency Sweep

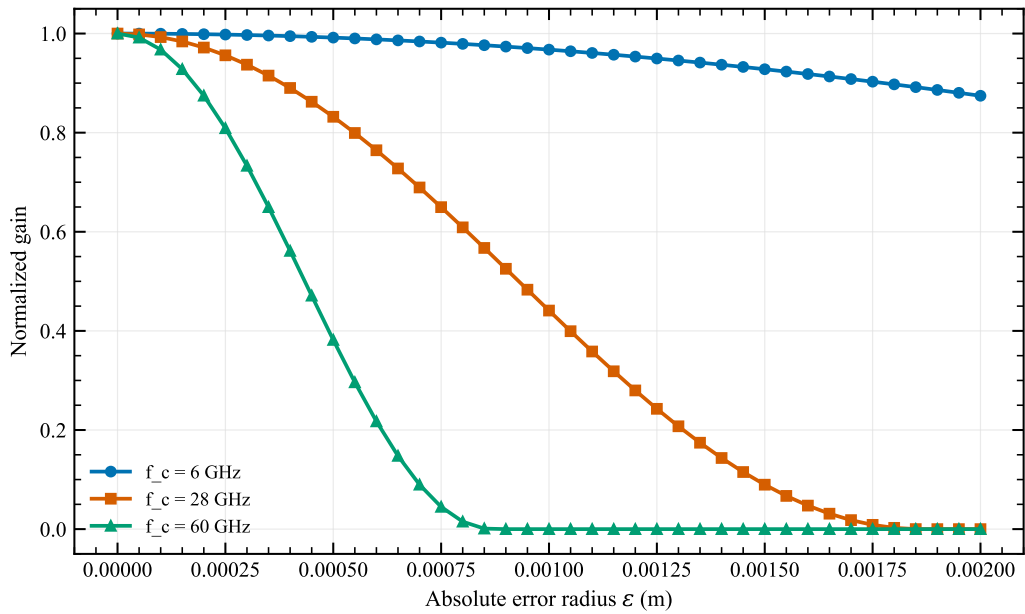


Figure 10: Frequency sweep under fixed absolute error levels. Higher carrier frequency makes the same physical pinching-position error much more destructive because the effective phase perturbation scales with k_0 .

At a representative absolute error of 1 mm, the normalized box-best-found gain is approximately 0.9676 at 6 GHz, 0.4411 at 28 GHz, and numerical zero at 60 GHz. The engineering implication is direct: if PASS is pushed toward higher mmWave or sub-THz operation, placement accuracy requirements tighten sharply.

B.3 Full Baseline Summary

Table 11: Expanded same-aperture baseline summary in raw array-gain value: gain-oriented metrics.

Placement	Nominal	Box-best-found	iid mean	Common-bias
Constructive aligned	1.7012	0.4435	1.3115	1.7011
Uniform aperture	1.6825	0.3886	1.2998	1.6825
Best-found nominal comparator	1.7299	0.4927	1.3336	1.7299

Table 12: Expanded same-aperture baseline summary in raw array-gain value: correlated robustness and auxiliary statistics.

Placement	Corr.-Gauss.	ξ std.	Span (m)
Constructive aligned	1.4430	0.01245	0.11824
Uniform aperture	1.4312	0.01244	0.11824
Best-found nominal comparator	1.4651	0.01250	0.11824

Two observations are worth emphasizing. The first is that the constructive aligned placement is not trivially equivalent to a uniform aperture, even though the aperture span is matched exactly. The second is that the ξ standard deviations of the three placements are numerically close, which suggests that robustness differences are not controlled by scalar spread alone. The detailed phase profile still matters.

C Reproducibility and Artifact Map

This appendix records the repository files, command entry points, and output artifacts that directly support the manuscript. The aim is not to repeat the methods section, but to leave a clear handoff trail from the report text back to the reproducible simulation artifacts.

C.1 Project Files Used in the Report

The report relies on the following repository files:

- `config.py`: fixed experiment configuration and random seed.
- `main.py`: top-level experiment orchestration.
- `design_delta_star.py`: constructive aligned placement generation.
- `pass_model.py`: PASS geometry, phase, and array-gain evaluation.
- `robustness.py`: deterministic lower bound, corner enumeration, and continuous-box best-found search.
- `theory_extensions.py`: weighted phase-variance mechanism and stochastic covariance predictor.
- `placement_baselines.py`: uniform-aperture and best-found nominal-comparator baselines.
- `results/config.json`: stored configuration snapshot and software versions.
- `results/data_main.csv`: main deterministic robustness curve.
- `results/data_box_validation.csv`: corner, split-mode, and box-best-found comparison.
- `results/data_error_scenarios.csv`: stochastic scenarios and quadratic approximations.
- `results/data_corr_length_sweep.csv`: fixed-error correlation-length sensitivity sweep.
- `results/data_baseline_summary.csv`: compact baseline summary.
- `results/data_baseline_curves.csv`: full baseline curves.
- `results/data_freq_sweep.csv`: frequency sweep.
- `results/data_neff_sweep.csv`: effective-index sweep.

C.2 Execution Contract

The report is based on the repository’s standard experiment command:

```
python main.py
```

Running this command from the repository root regenerates the result CSV files and figure assets in `results/`. The current report uses the fixed random seed 20260111, which is stored in both `config.py` and `results/config.json`. Because the seed is fixed, the numerical values cited in the report can be reproduced exactly from the current codebase, provided that the same software versions are available.

C.3 Software Versions Recorded by the Experiment Run

The stored configuration reports the following package versions:

- NumPy 2.3.5
- SciPy 1.16.3
- Matplotlib 3.10.6

C.4 Output Artifact Map

The paper-facing artifact map is as follows:

- `fig_pass_system_model_corrected.pdf`: analytical PASS system model used in Section 1.
- `fig_pass_system_model_3d_clean.pdf`: three-dimensional PASS system-model view used in Section 1.
- `fig_main.pdf`: main robustness curve used in Section 3.1.
- `fig_xi_distribution.pdf`: sensitivity distribution used in Section 2.2.
- `fig_error_scenarios.pdf`: stochastic scenario figure used in Section 3.2.
- `fig_corr_length_sweep.pdf`: correlation-length sensitivity figure used in Section 3.2.
- `fig_baseline_box.pdf`: same-aperture robustness baseline figure used in Section 3.3.
- `fig_baseline_tradeoff.pdf`: nominal-versus-robustness comparison used in Section 3.3.
- `fig_neff_sweep.pdf`: effective-index sensitivity figure used in Section 3.4.
- `fig_freq_sweep.pdf`: frequency-sensitivity appendix figure discussed alongside Section 3.4.

The system-model figures used in Section 1 and all generated curve figures are copied into the same LaTeX-facing figure folder so that the paper sources remain self-contained.

C.5 Claim-Discipline Checklist

To keep the paper-facing narrative synchronized with the actual evidence, the report follows the following wording rules:

- The continuous bounded-box search is always described as *box-best-found*.
- The constructive placement is described as a *constructive phase-aligned feasible placement*, not as a globally optimal or near-optimal design.
- The split-mode pattern is described as an empirical near-adversarial construction in the pre-null regime, not as an exact optimizer.
- The stochastic predictor is scoped to the studied design and to the tested range $0 \leq \varepsilon/\lambda \leq 0.10$.
- Comparative wording is restricted to the same-aperture baseline section and does not generalize beyond the tested placements.

These rules are not stylistic preferences. They are part of the reproducibility contract of the report, because they ensure that the prose stays aligned with the artifacts that actually justify the conclusions.